

Communication protocol

API development documentation protocol

Jason
20241215

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Protocol overview

This protocol is mainly used for data interaction with the algorithm layer of the general chassis. It allows users to display data and issue commands.。

Network communication services use WebSocket and use JSON string data format for communication.。

WebSocket belongs to the server-side push technology, which is essentially an application layer protocol that can realize full-duplex two-way communication of persistent connections. The HTTP protocol is a one-way communication protocol, and the server will return data only if the client initiates an HTTP request. The WebSocket protocol is a two-way communication protocol, and after establishing a connection, both the client and the server can actively send or receive data to each other. The advantages are as follows:

1. Do not send HTTP requests frequently, only need to send an HTTP request for websocket handshake, and then you can use the TCP connection to communicate through the websocket protocol, avoiding the waste of transmitting multiple HTTP headers.
2. The full-duplex channel breaks through the limitation that only the client can request the server when requesting HTTP, and the server can directly push messages to the client.
3. It can be passed while generating data, and there is no need to wait for data generation to be completed, which improves efficiency.
4. WebSocket data transmission is based on data frames and can be transmitted in fragments, without fear that the data is too large and the packet cannot be accommodated.

Detailed description of the protocol version

The version number uses the format `major.minor.patch` to distinguish the version of the protocol.

Major corresponds to the protocol added when the algorithm version is released.

Minor corresponds to major updates made during the major version, such as new fields or major changes to the field structure.

Patch correspond to `major.minor` versions with minor minor updates that do not affect the structure of the previous protocol, such as modifying field content or revising some description content.

Version 1.0.0

Infrastructure information and protocol addition of main data, some basic algorithm knowledge required for data analysis.

Common communication format

The computer communication program mainly has the following operations, which belongs to the general protocol format. All commands are formatted as follows, except that some fields differ in content. Specific differences can be found in the sections of the Specific Data Agreement section.

注册发布信息 / Sign up for posting information
发布具体数据 / Publish specific data
注销发布信息 / Log out of the publication information
订阅话题数据 / Subscribe to topic data
数据返回 / Data return
取消订阅话题数据 / Unsubscribe from topic data
调用服务接口 / Call the service interface

注册发布信息/Sign up for posting information

客户端需要往服务端发布数据前,需要向服务端注册该发布信息,发布的话题以及对应的数据类型. 这个只需要注册一次就好,一般在程序的初始化部分发布就好.

Before the client needs to publish data to the server, it needs to register the publication information, the published topic and the corresponding data type with the server. This only needs to be registered once, and it is usually released in the initialization part of the program.

```
{ "op": "advertise",  
  (optional) "id": <string>,  
  "topic": <string>,  
  "type": <string>  
}
```

其中具体释义如下 / The specific explanations are as follows

op 为定值,一定是 advertise

= op is a fixed value, Must be "advertise".

Id 为可选值,为当前注册的 id 号.用于区分不同的客户端连接

= Id is an optional value and is the currently registered id number. Used to distinguish between different client connections.

topic 为必填项,但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节

= topic is required, but the value varies depending on the function. The corresponding values of each function are described in the Specific Data Protocol section.

type 为必填项,但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节

= Type is required, but the value varies depending on the function. The corresponding values of each function are described in the Specific Data Protocol section.

发布具体数据/Publish specific data

只有在往服务端注册了发布信息之后才可以有效发布,否则服务端不会响应该命令.

Only after registering the publishing information with the server can it be validly published, otherwise the server will not respond to the command.

```
{ "op": "publish",
  (optional) "id": <string>,
  "topic": <string>,
  "msg": <json>
}
```

其中具体释义如下 / The specific explanations are as follows

op 为定值,一定是 publish

= op is a fixed value, Must be "publish"

id 为可选值,为当前注册的 id 号.用于区分不同的客户端连接

= id is an optional value and is the currently registered id number. Used to distinguish between different client connections.

topic 为必填项,但是根据不同的功能该值也不同.各个功能的具体值见 [具体数据协议](#) 章节

= topic is required, but the value varies depending on the function. The specific values of each function are described in the Specific Data Agreement section.

msg 为必填项,但是根据不同的功能该值也不同.各个功能的具体值见 [具体数据协议](#) 章节

= msg is required, but the value varies depending on the function. The specific values of each function are described in the Specific Data Agreement section.

注销发布信息/Log out of the publication information

客户端需要往服务端注销发布数据,也就是不需要再往服务端发布该话题数据.一般是在程序的结束进行发布. (但是由于结束的时候一般也断开连接了.服务端会自动注销掉.这个不是必须实现,但是为了逻辑规范还是添加为好)

The client needs to unregister the publishing data to the server, that is, there is no need to publish the topic data to the server. It is usually released at the end of the program. (However, it is generally disconnected at the end. The server will be automatically logged out. This is not mandatory, but it is better to add it for the sake of logical specs).

```
{ "op": "unadvertise",
  (optional) "id": <string>,
  "topic": <string>
}
```

其中具体释义如下 / The specific explanations are as follows

op 为定值,一定是 unadvertise

= op is a fixed value, Must be "unadvertised".

id 为可选值,为当前注册的 id 号.用于区分不同的客户端连接

= id is an optional value and is the currently registered id number. Used to distinguish between different client connections.

topic 为必填项,但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节

= topic is required, but the value varies depending on the function. The corresponding values of each function are described in the Specific Data Protocol section.

订阅话题数据/Subscribe to topic data

客户端需要从服务端订阅信息,也就是服务端往客户端发布一些信息.

The client needs to subscribe to the information from the server, that is, the server publishes some information to the client.

```
{ "op": "subscribe",
  (optional) "id": <string>,
  "topic": <string>,
  "type": <string>,
  (optional) "throttle_rate": <int>,
  (optional) "fragment_size": <int>,
  (optional) "compression": <string>
}
```

其中具体释义如下 / The specific explanations are as follows

op 为定值,一定是 subscribe

= op is a fixed value, Must be subscribe.

id 为可选值,为当前注册的 id 号.用于区分不同的客户端连接

= Id is an optional value. is the currently registered ID number. Used to distinguish between different client connections.

topic 为必填项,但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节

= topic is required item, but the value is also different according to different functions. The corresponding values for each function are shown in the specific data protocol section.

type 为必填项,但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节

= Type is required, but the value varies according to different functions. The corresponding values for each function are shown in the specific data protocol section.

Throttle_rate 为可选值(默认值为 0),该值单位为 ms.为该订阅消息之间的回传时间间隔.可以用于调整获取数据的频率.

= Throttle_rate is an optional value (the default is 0) in ms. This is the return interval between the subscription messages. Can be used to adjust the frequency of the acquired data.

fragment_size 为可选值,是返回 json 字符串的拆分阈值.也就是如果返回结果对应的字符串长度超过该值则进行拆包处理.拆包处理的数据返回格式具体见下面子章节 [数据返回](#)

= The fragment_size is the optional value and is the split threshold that returns the json string. That is, if the corresponding string length of the returned result exceeds the value, it is processed for unpacking. For the format of unpacking data, see the data return below.

Compression 为可选值(默认不进行压缩).设置值可为 png,这个功能用于地图跟图像的压缩

传输,使用 png 压缩方法进行数据压缩. 具体需要使用该功能的地方会标示出来,否则不要给服务端传输该值.

= **Compression Is optional (not compressed by default).** Set the value can be png, this function is used for the compression transmission of map and image, using png compression method for data compression. Specific need to use the function will be marked, otherwise do not transmit the value to the server.

数据返回/Data return

在客户端往服务端发布该话题对应的订阅信息后,服务端会往客户端返回该数据.频率非固定,有些数据会按照固频率,有些非固定频率,有些是偶尔返回.虽然在发布订阅消息的时候会有个 throttle_rate 的值设定,但是这个值也是相当于按最快这个频率处理,低于这个频率的还是按照原消息的频率发布.

After the client publishes the subscription information corresponding to the topic to the server, the server will return the data to the client. The frequency is not fixed, some data will be fixed, some are non-fixed, and some are returned occasionally. Although there will be a throttle_rate value setting when publishing the subscription message, but this value is equivalent to the fastest frequency processing, lower than this frequency or released according to the frequency of the original message.

整包数据接收/Whole package data receipt

如果订阅的信息数据量没有超过设定的 fragment_size

If the amount of information data for the subscription does not exceed the set fragment_size.

```
{ "op": "publish",
  "msg": <string>,
  "topic": <string>
}
```

其中具体释义如下 / The specific definitions are given as follows

op 为定值,一定是 advertise

= op is a fixed value, must be "advertise".

msg 为返回对应话题 topic 的具体数据,具体数据内容见 [具体数据协议](#) 章节

= The "msg" returns the specific data of the corresponding topic topic, and the specific data content is shown in the specific data protocol section.

topic 为返回数据对应的话题名称,但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节

= topic It returns the topic name corresponding to the data, but the value is also different according to different functions. The corresponding values for each function are shown in the specific data protocol section.

分包接收/Subcontracted receiving

如果订阅的信息数据量超过了设定的 fragment_size,会按照每包最大 fragment_size 的大小拆包返回。

If the amount of subscribed information data exceeds the set fragment_size, it will be unpacked according to the maximum size of fragment_size per packet and returned.

```
{ "op": "fragment",
  "id": <string>,
  "topic": <string>,
  "data": <string>,
  "num": <int>,
  "total": <int>
}
```

其中具体释义如下 / The specific definitions are given as follows

op 为定值,一定是 fragment. 标识当前数据包是拆分包的其中一部分。

= op is a fixed value, must be fragment. identify the current packet is part of the split package.

id 为识别号, 在这个模式下一定会返回,用于区分不同客户端的拆分数据,避免数据混乱。

= The id is the identification number, which will be returned in this mode to distinguish the split data of different clients to avoid data confusion.

total 为拆包的总包数, 总共多少个包构成一个完整的数据包。

= Total is the total number of unpacking, how many packages constitute a complete data package.

num 为当前包对应的顺序号, 标识当前数据包是拆分包中的第几个。

= num Indicates the sequence number of the current packet, which identifies the number of the current packet in the subpackage.

data 为拆分的部分数据内容. 不要单独解析这个 data 字段!!! 这个是一个不完整的字符串, 先接收完完整的数据信息后, 将 data 字段的数据按照顺序进行字符串合并。这个字符串也是 json 结构的。合并之后使用 json 解析后, 生成的数据就是上面 [整包数据接收](#) 中的数据内容,再按照对应的解析方法解析就好。

= The data is part of the split data content. Don't parse this data field separately!!! This is an incomplete string. After receiving the complete data information, the data of the data field is merged in the string in order. This string is also json-structured. After merging, using json analysis, the generated data is the data content in the whole package data reception, and then parse according to the corresponding parsing method.

取消订阅话题数据/Unsubscribe from topic data

当不希望从服务端获取对应话题的消息时,则发布该部分到服务端,服务端就不会再往客户端发布数据了。

When you do not want to get the message of the corresponding topic from the server, the part will be released to the server, and the server will not release the data to the client again.

```
{  "op": "unsubscribe",
  (optional) "id": <string>,
  "topic": <string>
}
```

其中具体释义如下 / The specific definitions are given as follows

op 为定值,一定是 unsubscribe

= op is a fixed value, must be "unsubscribe".

Id 为可选值,为当前注册的 id 号.用于区分不同的客户端连接.

= Id is an optional value, and it is the currently registered id number. To distinguish between the different client connections.

topic 为必填项,但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节.

= topic is required item, but the value is also different according to different functions. The corresponding values for each function are shown in the specific data protocol section.

调用请求服务接口/Call the request service interface

该模式为应答模式,也就是发布调用服务的请求后,服务端会发送该请求的响应数据返回给客户端,但是返回的时间不定.

This mode is the answer mode, that is, after publishing the request of the call service, the server will send the response data of the request and return it to the client, but the return time is uncertain.

```
{  "op": "call_service",
  (optional) "id": <string>,
  "service": <string>,
  (optional) "args": <list<json>>,
}
```

其中具体释义如下 / The specific definitions are given as follows

op 为定值,一定是 call_service

= op is a fixed value, must be "call_service".

Id 为可选值,为当前发布服务的 id 号.用于区分不同时刻或者不同客户端的请求

= Id is optional, and is the id number of the current publishing service. Reests to distinguish different moments or different clients.

service 为必填项,是指该调用服务的名称.但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节.

= service is required; it means the name of the calling service. But the value varies according to the function. The corresponding values for each function are shown in the specific data protocol section.

args 为需要调用时传递的参数.但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节.这个数据类型为 json 格式.

= The args is the parameter passed when the call is required. But the value varies according to the function. The corresponding values for each function are shown in the specific data

protocol section. This data type is in the json format.

服务返回数据/Service returns data

该部分为 [调用请求服务接口](#) 返回对应的数据应答数据.数据返回的响应时间不定.可以用 id 进行区分.

= This section returns the corresponding data response data for the call request service interface. The response time of the data returns is variable. It can be distinguished with the id.

```
{ "op": "service_response",
  (optional) "id": <string>,
  "service": <string>,
  (optional) "values": <list<json>>,
  "result": <boolean>
}
```

op 为定值,一定是 service_response.

= Op is a fixed value, must be "service_response".

id 为可选值,为当前对应请求服务的 id 号.用于区分不同时刻或者不同客户端的请求.

= id is an optional value and is the ID number of the current corresponding request service. Used to distinguish between requests at different times or different clients.

service 为必填项,是指该调用服务的名称.但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节.

= service is required; it means the name of the calling service. But the value varies according to the function. The corresponding values for each function are shown in the specific data protocol section.

values 为返回请求的具体结果.但是根据不同的功能该值也不同.各个功能的对应值见 [具体数据协议](#) 章节.

=values To return the specific result of the request. But the value varies according to the function. The corresponding values for each function are shown in the specific data protocol section.

result 为对应请求的执行结果,为 bool 值,如果为 true 则代表该服务请求正确执行了.如果返回为 false 则代表该服务请求出错了.会在 value 中返回对应的错误消息.

= result is the execution result of the corresponding request, is the pool value, and if true, the service request is executed correctly. If the return is for false, the error occurred for the service request. The corresponding error message is returned in the value.

具体数据协议/Specific data agreement

订阅类(客户端从服务端获取数据)/ Subscription class

(client gets data from server)

这部分是客户端想要从服务端获取数据信息,主要相关的通用协议部分参照 [订阅话题数据](#)
[数据返回](#) [取消订阅话题数据](#)

This part is that the client wants to obtain data information from the server, and the main related general protocol part refers to "subscribing to topic data", "data return", and "unsubscribing to topic data".

规则为先发布对应的订阅话题数据,之后服务端会往客户端发布数据.如果客户端不再希望服务端发布数据到客户端.则发布取消订阅话题数据的指令.

The rule is to publish the corresponding subscription topic data first, and then the server will release the data to the client. If the client no longer wants the server to publish the data to the client. Publish an order to cancel the subscription to the topic data.

订阅的数据回调频率是不同的.根据服务端设置相关.有下面几种可能:

= The frequency of the data callback for the subscription is different. Set up related according to the server. There are several possibilities as described below:

1. 定时回调 (固定频率) / Timed callback (fixed frequency)
2. 状态转变时回调 (状态发生变化时才会触发,时间不定) / Callback when state changes (triggered when state changes, time varies)
3. 有效时尽可能固定 (只有当数据更新时才会回放,尽可能按照固定频率,但有可能会延时) / Fixed as much as possible when valid (playback will only be made when the data is updated, as far as possible according to a fixed frequency, but there may be delays)

获取机器人坐标 (定时触发) / Obtain the robot

coordinates (timing trigger)

该指令为获取机器人在地图中的坐标信息 (x,y,theta)

= This instruction is to obtain the coordinates of the robot in the map (x, y, theta)

先发第一条指令订阅广播 (发送)

= First first instruction subscribe to broadcast (send)

<pre>{ "op": "subscribe", "id": "get_pose", "topic": "/robot_pose", "type": "geometry_msgs/Pose2D" }</pre>	op:表示消息类型/Indicates the message type. advertise:广播/broadcast, id:识别号+随机数 / Identification number + random number. topic:消息名字 / Message name, type:消息类型 / Message type ,
--	--

从服务端返回以下数据。（接收）

= Return the following data from the server side.(receive)

<pre>{ "topic": "/robot_pose", "msg": { "y": -0.047, "x": 0.009, "theta": 0.0114 }, "op": "publish" }</pre>	topic:消息名字 / Message name, msg:消息 / message, x: 全局坐标 X 轴位置（地图中），单位米 = X: Global coordinate X-axis position (in the map), in meters y: 全局坐标 Y 轴位置（地图中），单位米 = y: Global coordinate Y-axis position (in the map), in meters theta: 全局坐标中方向（地图中），单位弧度 = theta: Direction in global coordinates (in map), in radians
---	--

如果不需要订阅该消息时，发送以下指令：（发送）

= If not required to subscribe to this message, send the following instructions: (send)

<pre>{ "op": "unsubscribe", "id": "get_pose", "topic": "/robot_pose" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
--	---

获取雷达点云数据（定时触发）/Acquisition of radar point cloud data (timing trigger)

该指令为获取雷达数据在机器人坐标系的数据值 (x,y)

=This instruction is to obtain the data value of the radar data in the robot coordinate system (x, y)

先发第一条指令订阅广播（发送）

<pre>{ "topic": "/laser_data", "type": "yutong_assistance/point_array", "id": "get_laser", "op": "subscribe" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
--	---

从服务端返回以下数据。（接收）

= Return the following data from the server side.(receive)

<pre>{ "topic": "/laser_data", "msg": { "px": [-0.276, 0.399, 1.634, 6.780, 3.219, 0.940, -0.103, -1.092, -2.927, -6.910, -2.495, -0.965], "py": [-1.197, -1.205, -1.213, -0.415, 1.718, 1.741, 1.747, 1.756, 1.770, 0.340, -1.178, -1.191], "pt": [0, 0, 0] } }</pre>	topic:消息名字 / Message name msg:消息 / Message px: 雷达坐标对应 X 轴位置，单位米 = px: The radar coordinates correspond to the X-axis position in meters py: 雷达坐标对应 Y 轴位置，单位米 = py: The radar coordinates correspond to the Y-axis position, in meters Pt: 针对雷达点云数据无效,默认为 0 的数组 = Pt: invalid data for radar point cloud,An array that defaults to 0
--	--

<pre> }, "op": "publish" } </pre>	这里面 px, py 的长度应该是一样的。不过不固定。这个示例程序删除了很多点云方便操作而已。 = The length of px and py should be the same. It's not fixed, though. This example program removes a lot of point clouds for easy operation. 上述的 px,py 数组的值是基于机器人视角的,如果要转换到地图上的表达,需要跟机器人位置结合。参照“雷达点云转换到全局坐标系下”中讲解公式进行融合处理。 = The values of the above px,py array are based on the robot's perspective, and if you want to convert to the representation on the map, you need to combine it with the robot's position. Refer to the formula explained in "Radar point cloud conversion to global coordinate system" for fusion processing.
-----------------------------------	---

如果不需要订阅该消息时,发送以下指令:(发送)

= If you do not need to subscribe to the message, send the following instruction: (Send)

<pre> { "topic": "/laser_data", "type": "yutong_assistance/point_array", "id": "get_laser", "op": "unsubscribe" } </pre>	op:表示消息类型 / Indicates the message type, advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
--	---

获取导航规划路径(有效时尽可能固定)/Get navigation planning paths (fixed as possible when effective)

该指令为获取规划路径在地图中的表达 (x,y)

= This command is to get the representation of the planned path in the map (x,y)

先发第一条指令订阅广播(发送)

= First Instruction to Subscribe to Broadcast (Send)

<pre> { "topic": "/global_path", "type": "yutong_assistance/point_array", "id": "get_path", "op": "subscribe" } </pre>	op:表示消息类型 / Indicates the message type, advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
--	---

从服务端返回以下数据。(接收)

=The following data is returned from the server. (Receiving)

<pre> { "topic": "/global_path", "msg": { "px": [-5.199, -5.202, -5.198, -5.1948771774, -5.190, -5.186, -5.182], "py": [-3.299, -3.255, -3.230, -3.206, </pre>	topic:消息名字 / Message name msg:消息 / Message px: 路径在地图坐标中对应 X 轴位置, 单位米 =Px: The path corresponds to the X-axis position in meters in the map coordinates py: 路径在地图坐标中对应 Y 轴位置, 单位米
--	---

<pre>-3.181, -3.156, -3.132, -3.107, -3.083, -3.060, -3.036], "pt": [0, 0, 0] }, "op": "publish" }</pre>	=py: The path corresponds to the Y axis position in meters in the map coordinates pt: 无用预留 =pt: Useless reservation 这里面 px, py 的长度应该是一样的。不过不固定。这个示例程序删除了很多点云方便操作而已 = The length of px and py should be the same. It's not fixed, though. This example program removes a lot of point clouds for easy operation.
--	---

如果不需要订阅该消息时，发送以下指令：（发送）

= If you do not need to subscribe to the message, send the following instruction: (Send)

<pre>{ "topic": "/global_path", "type": "yutong_assistance/point_array", "id": "get_laser", "op": "unsubscribe" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

获取地图(有效时尽可能固定)/ Get a map (fixed as much as possible when valid)

该指令为获取地图的信息,地图以图片的形式进行发送.

= This command is to obtain the information of the map, and the map is sent in the form of a picture.

发送下面一条指令：（发送）

= Send the following instruction: (send)

<pre>{ "compression": "png", "topic": "/map", "fragment_size": 6000, "type": "nav_msgs/OccupancyGrid", "id": "get_map", "op": "subscribe" }</pre>	说明(explain): op:表示消息类型 / Indicates the message type. subscribe:订阅消息 / Subscribe to the message, id:识别号+随机数/ Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type, fragment_size: 拆包字符串长度阈值,如果数据太大的话, 会采用分包的方式进行拆分发送,建议值 6000 = fragment_size: Unpack string length threshold, if the data is too large, it will be split and sent by subpacket, the recommended value is 6000 compression:压缩方式, 采用 png 压缩 = compression: Compression method, using PNG compression
---	---

如果不需要订阅该消息时，发送以下指令：（发送）

= If you do not need to subscribe to the message, send the following instruction: (Send)

<pre>{</pre>	op:表示消息类型 / Indicates the message type.
--------------	--

<pre> "topic": "/map", "type": "nav_msgs/OccupancyGrid", "id": "get_map", "op": "unsubscribe" } </pre>	advertise:广播 / message, id: 识别号+随机数 / Identification number + random number, topic: 消息名字 / Message name, type: 消息类型 / Message type,
--	---

从网络端口接收到以下两种可能返回的数据格式:

= Received from the network port in the following two possible returned data formats:

整包接收（接收）/Whole-package receiving (receive)

<pre> { "topic": "/map", "msg": { "info": { "origin": { "position": { "y": -9.6, "x": -9.6 }, "orientation": { "y": 0, "x": 0, "z": 0, "w": 1 } }, "width": 384, "resolution": 0.05, "height": 384 }, "data": "iVBORwOKGgoAAAANSUheUgAAAYAAAAGAC AAAAACBrOpjAAAKQ0IEQVR4Ae3BAY4I2W1E0 RvcUYL7X0Igd0R4pseSIzb6N8xXbeYUel7MmiS zJsmsSTJrksyaJLMmyaxJMMuSzJoksybJrEkya5L MmiSzJsmsSTJrksyaJLMmyaxJMMuSzJoksybJrE kya5LMmiSzJsmsSTJrksyaJLMmyaxJMMuSzJoks ybJrEkya5LMmiSzJsmsSTJrksyaJLMmyaxJMMuS zJoksybJrEkya5LMmiSzJsms\\6s0f0rzlRtMh3J VzM\\kU7+BkyPTeMWATidTqgAnBVABT84+Tk nPzjBV1TgNP8mTTorTFYU3FcURHFfvEyYHpmG LL7azX908fcQpkemIYuvdV83P3XdF28XpkemIYs vdvNBFm8XpkemIYuvFeaDLN4uTI9MQxZf607z c1m8XZgemYYsvlaYD7J4uzA9Mg1ZfK3A\\Vfw </pre>	说明: topic: 消息名称 / Message name op: png msg: 地图信息 / cartographic information Info: 地图相关信息 / Map related information Origin/position 第一个像素值所代表的全局坐标值 / The global coordinate value represented by the first pixel value Origin/orientation 这个可忽略，默认 / This can be ignored, default width: 地图的宽 / The width of the map Height: 地图的高 / The height of the map Resolution: 地图的分辨率，一个像素代表实际大小，示例 0.05m 一个像素格 / The resolution of the map, one pixel represents the actual size, example 0.05m one pixel lattice. data: 数据（base64 编码），解码后应该是可以直接使用图像读取的工具库就可以直接解析数据显示出来了。 = Data: data (base64 encoding), decoding should be directly using the image reading tool library can directly parse the data display.
--	--

WbxemR6Yhi68V5oMs3i5Mj0xDfI\\rTvNzWb
xdmB6Zhiy+VmB+Lou3C9Mj05DF1wrzQRZvF6Z
HpiGLr3Wn+bks3i5Mj0xDfI8rMD+XxduF6ZFpy
OJr3XySxduF6ZFpyOJrhfkgi7cL0yPTkMXXCsZPZ
fF2YXpkGrL4WmE+yOLtwvTINGTtxtcJ8kMXbhe
mRacjia4X5Ilu3C9Mj05DF1wrzQRZvF6ZHpiGLL
xZ8UrxdmB6ZhizWQ5gemYYs1kOYHpmGLNZD
mB6ZhizWQ5gemYYs1kOYHpmGLNZDmB6Zhiz
WQ5gemYYs1kOYHpmGLNZDmB6ZhizWQ5ge
mYYs1kOYHpmGLNZDmB6ZhizWQ5gemYYs1k
OYHpmGLNZDmB6ZhizWQ5gemYYs1kOYHpm
GLNZDmB6ZhizWQ5gemYYs1kOYHpmGLNZDm
B6ZhizWQ5gemYYs1kOYHpmGLNZDmB6ZhizW
Q5gemYYs1kOYHpmGLNZDmB6Zhiz+Xu40WTf
JTxRnwvTINGTxWsFPOPIXhjT\\kMWRMD0yD
Vm8VpiGLI6E6ZFpyOK1wjRkcSRMj0xDFq8VpiG
LI2F6ZBqyeK0wDVkcCdMj05DFa4VpyOJImB6Z
hixeK0xDFkfC9Mg0ZPFaYRqyOBKmR6Yhi9cK05
DFkTA9Mg1ZvFaYhiyOhOmRacjitcl0ZHEkTI9M
QxavFaYhiyNhemQasnitMA1ZHAnTI9OQxWuF
acjISJgemYYsXitMQxZHwvTINGTxWmEasjgSpke
mIYvXCtOQxZEwPTINWbxWmIYsjoTpkWnl4rXC
NGRxJEyPTeMWrxWmIYsjYXpkGrJ4rTANWRwJ
0yPTkMVrhWnl4kiYHpmGLF4rTEMWR8L0yDR
k8VphGrI4EqZHpiGLV7mv+wLurKgwDVkcCdMj
05DFoeC3MQ1ZHAnTI9OQxaEwr5LFkTA9Mg1Z
HArzKlkcCdMj05DFoTCvksWRMD0yDVkcCvMq
WRwJ0yPTkMWhMK+SxZEwPTINWRwK8ypJcS
JMj0xDFofCvEsWJ8L0yDRkcSjMqyTFiTA9Mg1Z
HArzKklxIkyPTeMWWh8K8SxYnwvTINGRxKMyrJ
MWJMD0yDVkcCvMqSXEiTI9MQxaHwrxLFifC9
Mg0ZHEozKskxYkwPTINWRwK8ypJcSJMj0xDFo
fCvMoVxYkwPTINWRwK8ypJcSJMj0xDFofCvEs
WJ8L0yDRkcSjMu2RxlkyPTeMWWh8K8SIKcCNMj
05DFoTDvksWJMD0yDVkcCvMuWZwi0yPTkM
WhMO+SxYkwPTINWRwK8ypJcSJMj0xDFofCvE
sWJ8L0yDRkcSjMu2RxlkyPTeMWWh8K8SxYnwvT
INGRxKMy7ZHEiTI9MQxaHwrxKUpwl0yPTkM
WhMO+SxYkwPTINWRwK8y5ZnAjTI9OQxaEw7
5LFiTA9Mg1ZHArzLlmcCNMj05DFoTCvkhQnwv
TINGRxKMy7ZHEiTI9MQxaHwrxLFifC9Mg0ZHE
ozLtkcSJMj0xDFofCvMoVxYkwPTINWRwKvo5J\\
\\IB3gpMfnPyTkW+c\\MkXJ8L0yDRk8f\\hvvg
398V9ERUUYa6ggv\\hK5xOfvAVdWdFBTiLO+u
+7uu+4CYpuK8ovkKYHpmGLNZDmB6ZhizWQ5
gemYYs1kOYHpmGLNZDmB6ZhizWQ5gemYYs1

```

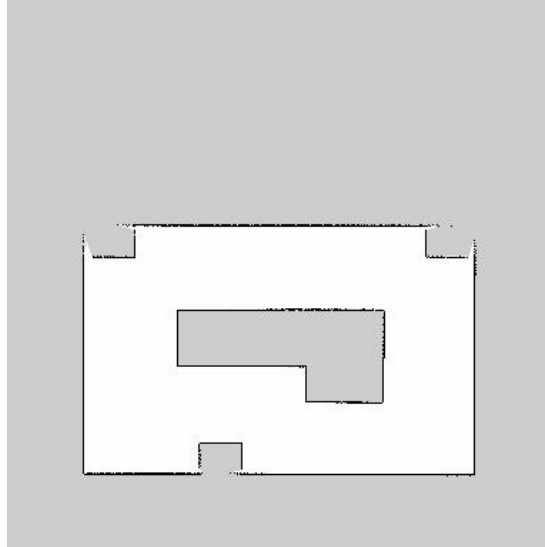
kOYHpmGLNZDmB6ZhizWQ5gemYYs1kOYHpm
GLNZDmB6ZhizWQ5gemYYs1kOYHpmGLNZDm
B6ZhizWQ5gemYYs1kOYHpmGLNZDmB6ZhizW
Q5gemYYs1kOYHpmGLNZDmB6ZhizWQ5gemY
Ys1kOYHpmGLNZDmB6ZhizWQ5gemYYs1kNg
WmQaslgPgWmRachiPYTpKwnIYj0EpkWmIYv1
EKZHpiGL9RCYFpmGLNZDmB6ZhizWQ2BaZBq
yWA9hemQaslgPYXpkGrJYD4FpkWnIYj2E6ZFpy
OJPwQ9O\\pPi+wgn\\+TkXzgLCEyLTEMWfw
qTTpNOp9P8Jc0fsvg2wllEEb6ibi64L+6LqPu6r\\
/uCMD0yDVn84eaTLL6NMJ9cEKZHpuPiDzefZP
FthPnkgjtNi0xDfNdZURbfRpiPLsL0yDRkATcfZF
thPno4k7TItOQBdx8IMW3EeajizA9Mg1ZcPNZF
t9GmI8uuOmRaci6r5vPsvg2wnx2BaZFpuO6r5v
Psvg2wnx23TTJtFw3v5DFtxHmsytMj0xD+rr5hS
y+jTC\\cN30yDRk3fxKFt9GmF9I0yPTcd38Shbf
RpjfRaYh6+ZXsvg2wvwuMg3XzS9I8W2E+V1kG
tL82sW3EeZ3kfINsvg2wvwuMr9JFt9GmN9F5jf
J4tsI87vI\\BYX38zN7yHz9S6+p5uvJ\\PFLoi6r
8DXfUE4676iCJP84AQq+Dtx+rohzV\\SfA2ZL5
S+7nRWmOQ\\c\\K\\OZ38kxOc\\OCECv5S
4QSc\\IMTJ\\g5L85+YuTP5kEJzhxghOn02
nSaUjqTqcTTGLICvNDYrjCpEnz1WTWJk1SWZN
klmTZNYkmTVJZk2SWZNk1iSZNUImTZJk2TWJ
Jk1SWZNklmTZNYkmTVJZk2SWZNk1iSZNUImT
ZJZk2TWJk1SWZNklmTZNYkmTVJZk2SWZNk1i
SZNUImTZJk2TWJk1SWZNklmTZNYkmTVJZk2
SWZNk1iSZNUImTZJk2TWJk1SWZNklmTZNYk
mTVJZk2SWZNk1iSZNUImTZJk2TWJk1SWZNk
ImTZNYkmTVJZk2SWZNk1iSZNUImTZJk2TWJ
k1SWZNklmTZNYkmTVJZk2SWZNk1iSZNUImTZ
JZk2TWJk1SWZNklmTZNYkmTVJZk2SWZNk1iS
ZNUImTZJk2TWJk1SWZNklmTZNYkmTVJZk2S
WZNk1iSZNUImTZJk2TWJk1SWZNklmTZNYk
mTVJZk2SWZNk1iSZNUImTZJk2TWJk1SWZNk
ImTZNYkmTVJZk2SWZNk1iSZNUImTZJk2TWJ
k1SWZNklmTZNYkmTVJZk2SWZNk1iSZNUImTZ
JZk2TWJk1SWZNklmTZNYkmTVJZk2SWZNk1iS
ZNUImTZJk2TWpP8Cm\\MBPYITy9QAAAAAS
UVORK5CYII="
    },
    "op": "png"
  }

```

其中 `data` 是包含的图像的数据。原始地图经过了图像 `png` 压缩-->`base64` 编码。所以解析的时候，是反过来的步骤，先把 `data` 进行 `base64` 解码，之后看编程语言使用图像库（应该有

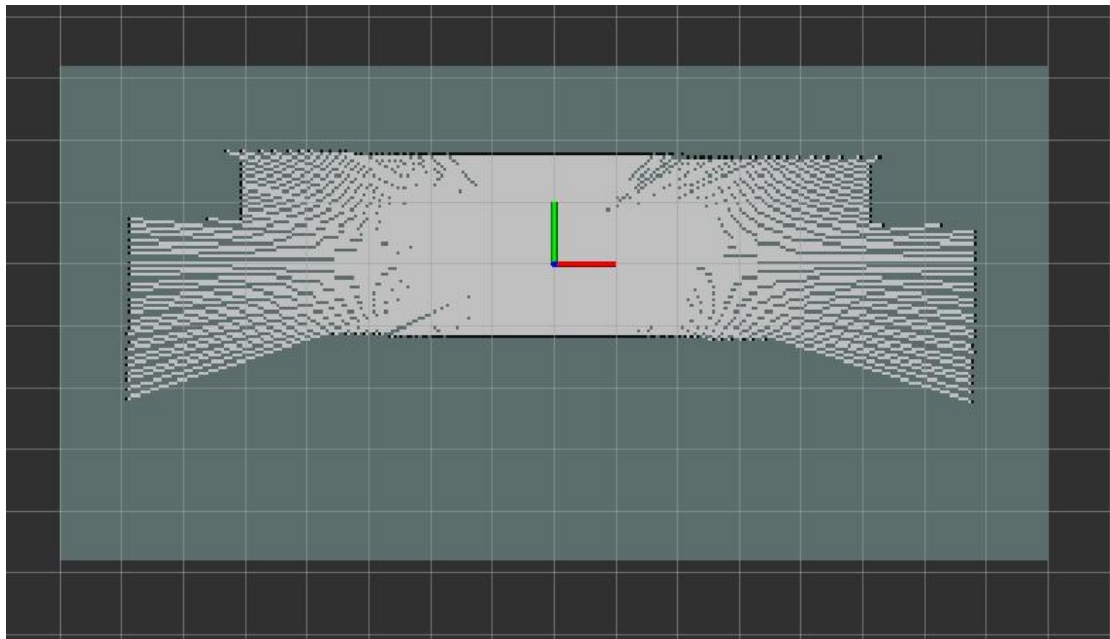
可以直接读取 png 压缩后的图片数据）可以直接显示数据。

= Where the data is the data of the included images. The original map underwent image png compression- -> base64 encoding. Therefore, when parsing, it is the reverse step, first data base64 decoding, and then see the programming language using the image library (there should be directly read png compressed image data) can directly display the data.



对应显示地图如上所示。（这个图片做过翻转显示处理，所以（0,0）像素坐标在左下角）根据 msg 中的 info 字段，可以得出，左下角的世界坐标是 (-9,6, -9,6)。图像的长宽分别是 384, 384, 一个像素栅格代表着 0.05m。

= The corresponding display map is shown above.(This picture has been flipped over, so (0,0) the pixel coordinates are in the lower left corner) According to the info field in msg, we can conclude that the world coordinate in the lower left corner is (-9,6, -9,6). The length and width of the image are 384,384, and a pixel grid represents 0.05m.



上图的长 height 为 160，宽 width 为 320。这个只是示例表达出地图的长宽表达。坐标原点（左下角）对应的世界坐标为(-8.0, -4.8)。这里面大的边框栅格是代表 1m 的长度。红绿色标记代表着世界坐标的原点（0,0）的位置。

= The long height in the figure above is 160 and the wide width is 320. This is only the example

expressing the broad expression of the map.

The corresponding world coordinate of the coordinate origin (the lower left corner) is (-8.0, -4.8). The large border grid in this area represents the length of 1m. The red-green mark represents the position of the origin (0,0) of the world coordinates.

分包接收 / Subcontracted receiving

从服务端返回以下数据。参见 [分包接收](#)

= The following data is returned from the server. See Subcontracted Reception

<pre>{ "total": 4, "num": 0, "data": "{xxxxx}", "id": 0, "topic": "/map", "op": "fragment" }</pre>	Op: fragment 标识当前数据包是拆分包的其中一部分 / Identifies that the current packet is a part of the split package Id: 识别号,这个会返回订阅时候设置的 id 号.用于区分不同客户端或者不同时间段的拆包信息 / Identification number, which returns the id number set during the subscription. Unpack information used to distinguish between different clients or different time periods Topic: 当前拆包的数据包对应的是哪个话题数据,按照上位机需求新增字段内容 / Which topic data corresponds to the current unpacking data package, according to the host requirements Total: 总共多少个包构成一个包 / How many packages in total make up a package Num: 标识当前数据包是拆分包中的第几个 / Identifying the current packet is the number one in the split package Data: 重要, 不要单独解析这个 data 字段。这个是一个不完整的字符串, 先接收完完整的数据信息后, 将 data 字段的数据按照顺序进行字符串合并。这个字符串也是 json 结构的。合并之后使用 json 解析后, 生成的数据就是上面整包发送的数据内容。后面步骤跟上面一样就好。 / Important, don't parse this data field separately. This is an incomplete string. After receiving the complete data information, the data of the data field is merged in the string in order. This string is also json-structured. After merging, using json resolution, the generated data is the data content sent by the whole package above. The next steps are just the same as the above.
--	---

机器人全局状态 / Robot global state

先发第一条指令订阅广播（发送）

= First first instruction subscribe to broadcast (send)

<pre>{</pre>	op:表示消息类型 / Indicates the message type.
--------------	---

API development documentation protocol

<pre>"topic": "/robot_status", "type": "yutong_assistance/RobotStatus", "id": "get_robot_status", "op": "subscribe" }</pre>	advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	--

从服务端返回以下数据。（接收）

= Return the following data from the server side.(receive)

<pre>{ "topic": "/robot_status", "msg": { "current_building_name": "tefa", "soft_estop": false, "battery": 0, "charger": 0, "nav_status": 0, "patrol_status": 0, "current_floor_name": "3", "current_goal_coordinate": { "y": 0.0, "x": 0.0, "theta": 0.0 }, "hard_estop": false, "velocity": [0.0, 0.0], "control_state": 30, "current_goal_name": "" }, "op": "publish" }</pre>	topic:消息名字 / Message name msg:消息 / message current_building_name(str) 当前地图对应的建筑名 / The building name corresponding to the current map current_floor_name(str) 当前地图对应的楼层名 / The floor name corresponding to the current map soft_estop(bool): 软急停,true 为急停中 / soft emergency stop, true is in an emergency stop state. hard_estop(bool): 硬件急停,true 为急停中 / hard emergency stop,true is in an emergency stop state. Battery (int): 电池电量的百分比 / The percentage of the battery charge. charger (int): 充电状态 0 未连接/1 成功充电/2 回充进行中/-1 回充失败.具体失败原因参见回充结果部分 / Charging status 0 not connected / 1 successful charging / 2 recharge in progress / -1 recharge failed. See the recharge result section for the specific reason of the failure. Nav_status(int): 当前导航状态 等待 600/运行中 601/取消 602/成功 603/失败 604。 具体可参见 状态码表 / Current navigation status is waiting for 600 / running 601 / cancel 602 / success 603 / failure 604. Please refer to the status code table for specific details. Patrol_status(int): 多点导航状态 / Multi-point navigation state Control_state(int): 当前程序状态 建图 20/定位导航 30/错误 99 / Current program status drawing 20 / Positioning navigation 30 / Error 99. Velocity (list): 数组形式,第一个为线速度(m/s),第二个为角速度(rad/s) / Array form, the first is linear velocity (m/s) and the second is angular velocity (rad/s) Current_goal_name(str): 当前导航目标点名称 (如果为空则是使用了坐标值的方式发布), 信息跟标记点位信息对应 / The name of the current navigation target point (if it is empty, the coordinate value is used), and the information corresponds to the mark point information. Current_goal_coordinate(dict): 当前导航点的具体坐标 / The specific coordinates of the current navigation point x: 地图中的坐标系 x 轴值,单位米 / Coordinate system x-axis value in the map, in meters y: 地图中的坐标系 y 轴值,单位米 / Coordinate frame y-axis
---	---

	values in the map, in meters theta: 地图中的的角度表达,单位弧度 / Angle expression in the map, unit radian
--	--

如果不需要订阅该消息时,发送以下指令:(发送)

= If not required to subscribe to this message, send the following instructions: (send)

{ "topic": "/robot_status", "type": "yutong_assistance/RobotStatus", "id": "get_robot_status", "op": "unsubscribe" }	op:表示消息类型 / Indicates the message type, advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	--

导航状态 (状态转变时)/Navigation state (when state transition)

该指令为获取涉及导航的信息,是运行中还是停止了,停止的原因是什么

= The instruction is to obtain the information involving navigation, is it running or stopped, what is the reason of the stop.

先发第一条指令订阅广播(发送)

= First first instruction subscribe to broadcast (send)

{ "topic": "/navi_status", "type": "actionlib_msgs/GoalStatus", "id": "get_map", "op": "subscribe" }	op:表示消息类型 / Indicates the message type, advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	--

从服务端返回以下数据。(接收)

=Return the following data from the server side.(receive)

{ "topic": "/navi_status", "msg": { "status": 3, "text": "Current goal is success", "goal_id": { "stamp": { "secs": 5976, "nsecs": 489000000 }, "id": "/move_base-4-5976.489000000" } }, "op": "publish"	topic:消息名字 / Message name msg:消息 / message Status(int8): 导航的状态信息 / State information for the navigation 0 开机初始化等待/0 Power-on initialization wait 1 目标点运行中/1 target point in running 2 目标点被用户取消/2 The target point is cancelled by the user 3 目标点成功到达/3 The target point was reached successfully 4 目标点导航失败/4 Target point navigation failed Text(str): 具体消息文本/Specific message text Goal_id 对应下发目标点的id号(目前可忽略)/The id number of the corresponding issue target point (currently negligible)
--	--

}	
---	--

如果不需要订阅该消息时，发送以下指令：（发送）

= If not required to subscribe to this message, send the following instructions: (send)

<pre>{ "topic": "/navi_status", "type": "actionlib_msgs/GoalStatus", "id": "get_map", "op": "unsubscribe" }</pre>	op:表示消息类型 / Indicates the message type, advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

激光检测障碍物区域/Laser detects the obstacle area

激光检测在指定周围范围内的障碍物的区域信息

= The laser detects the area information of obstacles within a specified surrounding range

该指令为获取机器人周边哪里检测到了障碍物并强制停止移动

= The command is to get where the obstacle is detected around the robot and force it to stop moving

先发第一条指令订阅广播（发送）

= First first instruction subscribe to broadcast (send)

<pre>{ "topic": "/obstacle_region", "type": "std_msgs/Int8", "id": "get_obstacle_region", "op": "subscribe" }</pre>	op:表示消息类型 / Indicates the message type, advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

从服务端返回以下数据。（接收）

= Return the following data from the server side.(receive)

<pre>{ "topic": "/obstacle_region", "msg": { "data": 2 }, "op": "publish" }</pre>	topic:消息名字 / Message name, msg:消息 / Message Data(int): 障碍物区域信息/Obstacle area information 0: 周围无障碍物/There are no obstacles around 1: 正右侧 90 度范围内有障碍物/There are obstacles within 90 degrees on the right right side 2: 正前方 90 度范围内有障碍物/There are obstacles within 90 degrees directly ahead 4: 正左方 90 度范围内有障碍物/There are obstacles within 90 degrees on the left
---	---

如果不需要订阅该消息时，发送以下指令：（发送）

= If not required to subscribe to this message, send the following instructions: (send)

<pre>{ "topic": "/obstacle_region",</pre>	op:表示消息类型 / Indicates the message type, advertise:广播 / message,
---	---

<pre> "type": "std_msgs/Int8", "id": "get_obstacle_region", "op": "unsubscribe" } </pre>	id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name, type:消息类型 / Message type,
--	---

地图匹配置信度/Map-matching of the confidence

scores

当前算法计算出来的地图跟激光雷达数据计算出的匹配置信度. 值范围为 0~1.0. 对应百分比形式.

= The map calculated by the current algorithm matches the confidence degree calculated by the lidar data. Values range from 0 to 1.0. Corresponding percentage form.

先发第一条指令订阅广播（发送）

= First first instruction subscribe to broadcast (send)

<pre> { "topic": "/localization_confidence", "type": "std_msgs/Float64", "id": "get_localization_confidence", "op": "subscribe" } </pre>	op:表示消息类型 / Indicates the message type , advertise:广播 / message, id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name, type:消息类型 / Message type,
--	--

从服务端返回以下数据。（接收）

= Return the following data from the server side.(receive)

<pre> { "topic": "/localization_confidence", "msg": { "data": 0.6 }, "op": "publish" } </pre>	topic:消息名字/Message name msg:消息/Message Data(double): 当前激光雷达跟地图匹配对应的置信度.取值范围 0 ~ 1.0. 对应百分比形式 / The confidence level of the current lidar and map match. Value range from 0 to 1.0. Corresponding percentage form
---	--

如果不需要订阅该消息时，发送以下指令：（发送）

= If not required to subscribe to this message, send the following instructions: (send)

<pre> { "topic": "/localization_confidence", "type": "std_msgs/Float64", "id": "get_localization_confidence", "op": "unsubscribe" } </pre>	op:表示消息类型 / Indicates the message type , advertise:广播 / message, id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name, type:消息类型 / Message type,
--	--

地图匹配安全防护阈值/The map matches the security protection threshold

返回当前地图匹配触发安全防护的阈值。

= Returns the threshold where the current map match triggers security protection.

先发第一条指令订阅广播（发送）

= First first instruction subscribe to broadcast (send)

<pre>{ "topic": "/laser_safety_controller/confidence_threshold", "type": "std_msgs/Float64", "id": "get_match_threshold", "op": "subscribe" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

从服务端返回以下数据。（接收）

= Return the following data from the server side.(receive)

<pre>{ "topic": "/laser_safety_controller/confidence_threshold", "msg": { "data": 0.6 }, "op": "publish" }</pre>	topic:消息名字/Message name msg:消息/Message Data(double): 当前激光雷达跟地图匹配对应的置信度.取值范围 0 ~ 1.0. 对应百分比形式 / The confidence level of the current lidar and map match. Value range from 0 to 1.0. Corresponding percentage form
--	---

如果不需要订阅该消息时，发送以下指令：（发送）

= If not required to subscribe to this message, send the following instructions: (send)

<pre>{ "topic": "/laser_safety_controller/confidence_threshold", "type": "std_msgs/Float64", "id": "get_match_threshold", "op": "unsubscribe" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

主动推送通知/Active push notification

主动推送一些系统通知.具体通知内容下面一些

= **Actively push some system notifications. Specific notice content has some below**

- 地图匹配阈值触发跟激光雷达数据无效触发警示 / Map matching threshold trigger and lidar data invalid trigger warning

- 底盘传感器异常 / Chassis sensor is abnormal

先发第一条指令订阅广播（发送）

= **First first instruction subscribe to broadcast (send)**

<pre>{ "topic": "/notification", "type": "rosgraph_msgs/Log", "id": "get_/notification", "op": "subscribe" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
--	---

从服务端返回以下数据。（接收）

= **Return the following data from the server side.(receive)**

<pre>{ "topic": "/notification", "msg": { "function": "激光雷达", "name": "system_monitor", "level": 16, "topics": ["laser"], "header": { "stamp": { "secs": 1627005599, "nsecs": 23596048 }, "frame_id": "", "seq": 2 }, "file": "", "msg": "激光雷达未连接", "line": 0 }, "op": "publish" }</pre>	topic:消息名字/Message name msg:消息/Message Name(str): 主动提示的大类.具体是那些功能模块推送的 / Big category of the active prompt. Specifically, those function module push Level(int): 事件等级 1: 调试等级 2: 标准事件 4: 警告事件 8: 错误事件 16: 致命事件 0: 提示事件 = Level (int): Event Level 1: Debug Level 2: Standard Event 4: Warning Event 8: error Event 16: Fatal Event 0: prompt event Topics(array): 出事列表.目前底盘传感器中返回各个传感器名称 / Accident list. Individual sensor names are currently returned in the chassis sensor Header: Stamp: 时间戳信息.(自 1970.01.01 所经过秒数) / Time-stamp information. (Seconds since 1970.01.01) Secs (int): 秒 / S Nsecs(int): 纳秒 / NS (nanosecond) Msg: 具体消息内容 / Specific information content 主动通知信息具体释义内容可以参见附表 主动通知内容以及字段内容 = The specific definition of the active notice information can be found in the attached table for the active notice content and the field content
--	--

如果不需要订阅该消息时，发送以下指令：（发送）

= If not required to subscribe to this message, send the following instructions: (send)

<pre>{ "topic": "/notification", "type": "rosgraph_msgs/Log", "id": "get_/notification", "op": "unsubscribe" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
--	---

底盘传感器数据/Chassis sensor data

底盘上搭载的传感器信息如电池电量,电压,硬急停,充放电状态,充电桩连接状态。

= The sensor information carried on the chassis, such as battery power, voltage, hard emergency stop, charge and discharge state, and charging pile connection state.

充放电状态跟充电桩连接状态的区别是,充放电状态是指电池内部实际充放电状态,机器在充电桩上充电,但是电池内部过充保护触发的时候,充电状态还是会变成未充电。

= The difference between the charge and discharge state and the connection state of the charging pile is that the charge and discharge state refers to the actual charge and discharge state inside the battery. The machine is charging on the charging pile, but when the overcharging protection inside the battery is triggered, the charging state will still become uncharged.

充电桩连接状态是机器人是否在充电桩上,不管机器人电池内部是否实际在充放电。

= The connection state of the charging pile is whether the robot is on the charging pile, regardless of whether the robot battery is actually charged and discharged.

先发第一条指令订阅广播（发送）

= First first instruction subscribe to broadcast (send)

<pre>{ "topic": "/mobile_base/sensors/core", "type": "kobuki_msgs/SensorState", "id": "get_sensors_core", "op": "subscribe" }</pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

从服务端返回以下数据。（接收）

= Return the following data from the server side.(receive)

<pre>{ "topic": "/mobile_base/sensors/core", "msg": { "current": "AAA=", "right_encoder": 1503, "analog_input": [0, 39319, 0, 0], "bottom": [0, 0, 0], "battery": 97, "charger": 1, </pre>	topic:消息名字/Message name msg:消息/Message Header: Stamp: 时间戳信息.(自 1970.01.01 所经过秒数)/Time-stamp information. (Seconds since 1970.01.01) Secs (int): 秒 / S Nsecs(int): 纳秒 / NS (nanosecond) battery (int): 电池电量百分比. / Battery charge percentage. charger (int): 充电桩连接状态(机器人是否在充电桩上) 0 不
--	--

<pre> "over_current": 1, "left_pwm": -29, "buttons": 0, "header": { "stamp": { "secs": 1627033096, "nsecs": 700239558 }, "frame_id": "", "seq": 279096 }, "right_pwm": 25, "wheel_drop": 0, "bumper": 0, "time_stamp": 27582, "digital_input": 0, "left_encoder": 63529, "cliff": 0 }, "op": "publish" } </pre>	在充电桩上 / 1 在充电桩上 = charger (int): charging pile connection status (whether the robot is on the charging pile) 0 is not on the charging pile / 1 is on the charging pile. over_current (int): 充放电状态 0 未充电 / 1 充电 = over_current (int): charge / discharge state 0 not charged / 1 charge. Buttons (int): 硬件急停状态 0 急停未触发 / 1 急停触发 = Hardware emergency stop state 0 emergency stop not triggered / 1 emergency stop triggered. Analog_input (array): 超声波数据数组,单位为 mm. 目前只有数组第二个有效为正中间超声波 = Analog_input (array): array of ultrasound data in mm. Currently only the second array is positive middle ultrasound.
---	---

如果不需要订阅该消息时，发送以下指令：（发送）

= If not required to subscribe to this message, send the following instructions: (send)

<pre> { "topic": "/mobile_base/sensors/core", "type": "kobuki_msgs/SensorState", "id": "get_sensors_core", "op": "unsubscribe" } </pre>	op:表示消息类型 / Indicates the message type. advertise:广播 / message, id:识别号+随机数 / Identification number + random number, topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

发布类(客户端往服务端发布具体数据)/Release class (the client releases specific data to the server side)

该部分为客户端希望往服务端发布信息的功能. 对应的通用协议格式有 [注册发布信息](#) [发布具体数据](#) [注销发布信息](#)。

= This section is the function of the client wishing to publish information to the server. The corresponding common protocol formats are "registration and release information", "release of specific data", and "cancellation of release information".

注册发布信息只需要在初始化过程发布一次就可以了. 之后就可以一直发布具体数据内容。

= Registered release information only needs to be published once during the initialization process. After that, you can release specific data content all the time.

下面发布坐标相关的数据都是基于地图的坐标系下的表达.如果命令中包含了 header 字段的信息,则需要设置其子字段的 frame_id 为 map

= The coordinates-related data published below are based on the representation of the map's coordinate system. If the command contains information in the header field, you need to set the frame_id of its subfields to map.

速度控制命令/Speed control command

通过发布对应的控制速度来控制机器人,为差分驱动方式(线速度, 角速度).

= The robot is controlled by issuing the corresponding control speed, which is a differential drive mode (linear speed, angular velocity).

发布一次速度在算法层会持续响应 0.6 秒.最后停止时发一次全为零的速度就好.

= A single release speed lasts for 0.6 seconds. Finally stop with a full zero speed.

不要一直发布为零的速度,算法层会认为用户持续用手动遥控模式,导致导航无响应.

= Do not always publish zero speed, the algorithm layer will think that the user continues to use manual remote control mode, resulting in no navigation response.

先广播 (发送) /Broadcast first (send)

<pre>{ "op": "advertise", "id": "velocity_control", "topic": "/cmd_vel_mux/input/teleop", "type": "geometry_msgs/Twist" }</pre>	说明(explain): topic:消息名字 / Message name type:消息类型 / Message type
---	--

往"/cmd_vel_mux/input/teleop"消息上发送数据。循环发送 (发送)

= Send data to the "/cmd_vel_mux/input/teleop" message. Circular Send (Send)

<pre>{ "op": "publish", "topic": "/cmd_vel_mux/input/teleop", "id": "velocity_control", "msg": { "angular": { "z": 0 }, "linear": { "x": 0.2 } } }</pre>	说明(explain): msg:消息 / message angular:角速度, 单位 弧度每秒 rad/s = angular: Angle speed, rad / s per second z:0 // 旋转速度 = Z: 0 // rotation speed linear:线速度, 单位 米每秒 m/s = linear: linear velocity, m/s in meters per second x:0.2 // 前进速度 = x:0.2 // Forward speed
--	--

停止发布 subscribe (发送)

<pre>{ "op": "unadvertise", "id": "velocity_control", "topic": "/cmd_vel_mux/input/teleop"</pre>	说明(explain): op:表示消息类型 / Indicates the message type. advertise:广播 / Stop broadcasting, id:识别号+随机数 / Identification number + random number,
--	--

}	topic:消息名字 / Message name,
---	----------------------------

扩展的速度控制命令大概如下:

= The extended speed control command is roughly as follows:

- 直线前进需设置 msg->linear->x 为正值, 单位为 m/s; msg->angular->z 为 0
= msg->linear->x is set to a positive value for linear advance.The unit is m/s; msg->angular->z is 0.
- 后退需设置 msg->linear->x 为负值, 单位为 m/s; msg->angular->z 为 0
= msg->linear->x is negative when backing up.The unit is m/s; msg->angular->z is 0.
- 原地顺时针旋转需设置 msg->linear->x 为 0; msg->angular->z 为正值, 单位 rad/s
= For clockwise rotation in place, msg->linear->x is set to 0;msg->angular->z is a positive value in rad/s
- 原地逆时针旋转需设置 msg->linear->x 为 0; msg->angular->z 为负值, 单位 rad/s
= For counterclockwise rotation, msg->linear->x is set to 0;msg->angular->z is a negative value in rad/s
- 停止需设置 msg->linear->x 为 0; msg->angular->z 为 0
= To stop, set msg->linear->x to 0;msg->angular->z is 0

取消目标点导航 /Cancel the target point navigation

该指令的主要功能是取消目标点导航。

= The main function of this instruction is to cancel the target point navigation.

先广播 (发送) /Broadcast first (send)

{ "op": "advertise", "id": "cancel_goal", "topic": "/move_base/cancel", "type": "actionlib_msgs/GoalID" }	说明(explain): op:表示操作 / Indicates the operation advertise:广播 / Broadcast id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name, type:消息类型 / Message type,
--	--

往"/move_base/cancel"消息上发送数据。(发送)

{ "op": "publish", "topic": "/move_base/cancel", "id": "cancel_goal", "msg": { "stamp": "", "id": "" } }	说明(explain): op:发布 / publish topic:消息名字 / Message name id:识别号+随机数 / Identification number + random number , msg:消息 / Message stamp:时间戳 (为空即可) / Time stamp (just empty) id:目标点 id (为空即可) / Target point id (just empty)
--	---

停止发布 subscribe (发送) /Stop publishing the subscribe (send)

{	说明(explain):
---	--------------

<pre>"op": "unadvertise", "id": "cancel_goal", "topic": "/move_base/cancel" }</pre>	op:表示消息类型 / Indicates the message type advertise:停止广播 / Stop broadcasting id:识别号+随机数 / Identification number + random number , topic:消息名字 /Message name
---	---

软急停/Soft and urgent stop

该指令的主要功能是用于启动关闭软件设置急停功能

= The main function of this instruction is to start the emergency stop function

先广播 (发送) /Broadcast first (send)

<pre>{ "topic": "/soft_stop", "type": "std_msgs/Bool", "id": "set_estop", "op": "advertise" }</pre>	说明(explain): op:表示操作 / Indicates the operation advertise:广播 / Broadcast id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name, type:消息类型 / Message type,
---	---

往"/soft_stop"消息上发送数据。(发送)

<pre>{ "topic": "/soft_stop", "msg": { "data": true }, "id": "set_estop", "op": "publish" }</pre>	说明(explain): msg:消息 / Message Data(bool): 设置为 true 为开启软件急停功能.false 为关闭 = Data (pool): Set to true to enable the software emergency stop function. false is closed
---	---

停止发布 subscribe (发送)

<pre>{ "topic": "/soft_stop", "id": "set_estop", "op": "unadvertise" }</pre>	说明(explain): op:表示消息类型 /Indicates the message type advertise:停止广播 / Stop broadcasting id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name
--	---

当前机器人位置添加为标记点/The current robot position is added as a marked point

该指令的主要功能是用于将当前机器人所在位置添加到标记点列表中

= The main function of this instruction is to add the current location of the robot to the list of marked points.

先广播（发送） / Broadcast first (send)

<pre>{ "topic": "/insert_current_pose_marker", "type": "yutong_assistance/InsertPose", "id": "insert_current_pose_marker", "op": "advertise" }</pre>	说明(explain): op:表示操作 / Indicates the operation advertise:广播 / Broadcast id:识别号+随机数 / Identification number + random number topic:消息名字 / Message name type:消息类型 / Message type
--	---

往"/insert_current_pose_marker"消息上发送数据。（发送）

= Send data to the /insert_current_pose_marker message. (Send)

<pre>{ "topic": "/insert_current_pose_marker", "msg": { "time_out": 0, "behavior_code": 11, "name": "csyt2", "rest_time": 11 }, "id": "insert_current_pose_marker", "op": "publish" }</pre>	说明(explain): msg:消息/ Message Behavior_code(int): 当前标记点的属性,目前 11 为回充点 = Behavior_code (int): the property of the current marker point, 11 is the recharge point Time_out (float): 当前点对应的执行超时时间.单位秒.在规定的时间内如果该点还没有到达则取消该点,执行下一个点位.设置为-1 代表不设置超时时间.一直等待直到该点完成 = Time_out (float): The execution timeout corresponding to the current point. In seconds. If the point has not been reached within the stipulated time, cancel the point and execute the next point. Set to -1 to indicate that no timeout is set. Wait until the point is complete. Name (str): 当前点位的对应名称,该名称需唯一.如果不唯一的话其中一个会被覆盖 = Name (str): The corresponding name of the current point, which must be unique. If not the only one, one of them will be overwritten Rest_time (float) 当前点位的休息时间.到达该点位后等待多少秒后再继续进行去下一个点位 = Rest_time (float) The rest time of the current point. How many seconds to wait after reaching that point before proceeding to the next point
---	---

停止发布 subscribe（发送）/Stop publishing the subscribe (send)

<pre>{ "topic": "/insert_current_pose_marker", "id": "insert_current_pose_marker", "op": "unadvertise" }</pre>	说明(explain): op:表示消息类型 /Indicates the message type advertise:停止广播 / Stop broadcasting id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name
--	---

立马返回地图/Return to the map immediately

该指令的主要功能是用于立马触发地图数据回调的功能

= The main function of this instruction is used to trigger the map data callback immediately

先广播（发送） / Broadcast first (send)

<pre>{ "topic": "/get_current_map", "type": "std_msgs/Bool", "id": "get_current_map", "op": "advertise" }</pre>	说明(explain): op:表示操作 / Indicates the operation advertise:广播 / Broadcast id:识别号+随机数 / Identification number + random number topic:消息名字 / Message name type:消息类型 / Message type
---	---

往"/get_current_map"消息上发送数据。（发送）

<pre>{ "topic": "/get_current_map", "msg": { "data": true }, "id": "get_current_map", "op": "publish" }</pre>	说明(explain): msg:消息 / Message Data(bool): 设置为 true 为触发地图数据的立马回调 /Set to true to trigger the immediate callback of map data
---	---

停止发布 subscribe（发送）

<pre>{ "topic": "/get_current_map", "id": "get_current_map", "op": "unadvertise" }</pre>	说明(explain): op:表示消息类型 /Indicates the message type advertise:停止广播 / Stop broadcasting id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name
--	---

激光检测停止移动/Laser detection stops the movement

该指令的主要功能是用于启动关闭激光检测到障碍物时停止手动操控的功能

= The main function of this instruction is to stop the manual control when the laser detects an obstacle

先广播 (发送) /Broadcast first (send)

<pre>{ "topic": "/node_manager/laser_safety_controller", "type": "std_msgs/Bool", "id": "set_laser_detection", "op": "advertise" }</pre>	说明(explain): op:表示操作 / Indicates the operation advertise:广播 / Broadcast id:识别号+随机数 / Identification number + random number topic:消息名字 / Message name type:消息类型 / Message type
--	---

往"/node_manager/laser_safety_controller"消息上发送数据。(发送)

= Send data to the /node_manager/laser_safety_controller message. (Send)

<pre>{ "topic": "/node_manager/laser_safety_controller", "msg": { "data": true }, "id": "set_laser_detection", "op": "publish" }</pre>	说明(explain): msg:消息 / Message Data(bool): 设置为 true 为开启检测到周围障碍物停止移动功能, false 为关闭检测到周围障碍物停止移动功能 / Set to true to turn on the stop movement function when surrounding obstacles are detected. false turns off the stop movement function when surrounding obstacles are detected.
--	--

停止发布 subscribe (发送) /Stop publishing the subscribe (send)

<pre>{ "topic": "/node_manager/laser_safety_controller", "id": "set_laser_detection", "op": "unadvertise" }</pre>	说明(explain): op:表示消息类型 /Indicates the message type advertise:停止广播 / Stop broadcasting id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name
---	---

激光检测距离/Laser detection distance

该指令的主要功能是用于设置激光检测障碍物的距离值。距离值最低为 0.3 米.低于 0.3 米会被设置为默认最低值 0.3m。

= The main function of this instruction is to set the distance value of the laser detection obstacle. The lowest distance value was 0.3 m. Below 0.3 m is set to the default lowest value of 0.3m.

先广播 (发送) /Broadcast first (send)

<pre>{ "topic": "/node_manager/laser_safety_range", "type": "std_msgs/Float64", "id": "set_laser_safety_range", "op": "advertise" }</pre>	说明(explain): op:表示操作 / Indicates the operation advertise:广播 / Broadcast id:识别号+随机数 / Identification number + random number topic:消息名字 / Message name type:消息类型 / Message type
---	---

}	
---	--

往"/node_manager/laser_safety_range"消息上发送数据。(发送)

= Send data to the /node_manager/laser_safety_range message. (Send)

<pre>{ "topic": "/node_manager/laser_safety_range", "msg": { "data": 0.5 }, "id": "set_laser_safety_range", "op": "publish" }</pre>	说明(explain): msg:消息 / Message Data(double): 激光检测停止移动的检测阈值，单位米。在此距离范围内的障碍物检测到则会停止机器人的移动。 = Laser detection stops moving the detection threshold, per m. In this distance range, the robot will stop moving.
---	--

停止发布 subscribe (发送) /Stop publishing the subscribe (send)

<pre>{ "topic": "/node_manager/laser_safety_range", "id": "set_laser_safety_range", "op": "unadvertise" }</pre>	说明(explain): op:表示消息类型 /Indicates the message type advertise:停止广播 / Stop broadcasting id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name
---	---

地图匹配安全防护阈值调节/The map matches the security protection threshold adjustment

该指令的主要功能是用于地图匹配阈值来触发安全防护.在导航过程中,如果地图匹配的阈值低于该值则会触发安全防护行为,底盘会原地旋转后再停止运动(此时手动遥控无效).直到导航结束才会释放. 默认值为 0.8 (80%)

= The main function of this instruction is to map match thresholds to trigger security protection. During navigation, if the matching threshold of the map is lower than this value, the safety protection behavior will be triggered, and the chassis will rotate in place and then stop moving (the manual remote control is invalid at this time). Is not released until the navigation is over. Default value is 0.8 (80%).

先广播 (发送) /Broadcast first (send)

<pre>{ "topic": "/laser_safety_controller/setting_confidence_threshold", "type": "std_msgs/Float64", "id": "set_safety_match_threshold", "op": "advertise" }</pre>	说明(explain): op:表示操作 / Indicates the operation advertise:广播 / Broadcast id:识别号+随机数 / Identification number + random number topic:消息名字 / Message name type:消息类型 / Message type
--	---

往"/laser_safety_controller/confidence_threshold"消息上发送数据。（发送）

= Send data to the /laser_safety_controller/confidence_threshold message. (Send)

<pre>{ "topic": "/laser_safety_controller/setting_confidence_t hreshold", "msg": { "data": 0.5 }, "id": "set_safety_match_threshold", "op": "publish" }</pre>	说明(explain): msg:消息 / Message Data(double): 地图匹配对应触发安全防护的检测阈值。当此阈值低于地图匹配的阈值时,会触发安全防护的行为. 此阈值范围为 0~1.0 对应为百分比. = Map matching corresponds to the detection threshold of triggering safety protection. This threshold is below the threshold for map matching. This threshold range from 0 to 1.0 corresponds to the percentage.
---	---

停止发布 subscribe（发送）/Stop publishing the subscribe (send)

<pre>{ "topic": "/laser_safety_controller/setting_confidence_t hreshold", "id": "set_safety_match_threshold", "op": "unadvertise" }</pre>	说明(explain): op:表示消息类型 /Indicates the message type advertise:停止广播 / Stop broadcasting id:识别号+随机数 / Identification number + random number , topic:消息名字 / Message name
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服务类(客户端往服务端发布数据,返回对应结果)/Service class (client publishes data to server and returns corresponding result)

该部分为请求应答模式.客户端往服务端发送请求信息,服务端执行完对应操作之后返回应答信息给客户端. 相关通用协议部分为 [调用请求服务接口](#) [服务返回数据](#)

= This section is the request reply mode. The client sends the request information to the server, and the server returns the reply information to the client after performing the corresponding operation. The relevant common protocol sections are "Call Request Service Interface" and "Service Return Data".

程序控制框架 /Program control framework

该指令的主要功能是通过指令来进行建图,定位导航,保存地图,切换地图。

= The main function of this instruction is to build the map, locate the navigation, save the map, and switch the map.

发布该服务对应的请求（发布）

= The request corresponding to publishing the service (publishing)

<pre>{ "args": { "floor_num": "2", "cmd": 0, "args": 0, "building_name": "csst" }, "id": "service_node_manager_control", "service": "/node_manager_control", "op": "call_service" }</pre>	说明(explain): Op(str): call_service 标示该操作为请求服务 = Op(str): "call_service" indicates that the operation is a request service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 = id(str): Used to set the ID number corresponding to the service. It can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 /Input parameters corresponding to the service request Json 解析后的数据如下 /The parsed data are presented as follows Building_name(str): 当前地图的建筑名 /The building name of the current map Floor_num(str): 当前地图的楼层名 /Floor name of the current map Cmd(int): 当前对应的操作 /The current corresponding action 0: 开启建图(定位导航模式可用) /Open the drawing (positioning and navigation mode is available) 1: 继续建图 (预留) /Continue to build the drawing (reserved) 2: 重启建图(预留) /Restart the construction drawing (reserved) 3: 保存地图及相应信息 (任意模式都可使用) /Save the map and the corresponding information (available in any mode) 4: 开启定位导航 (建图模式可用) /Turn on location navigation (mapping mode is available) 7: 切换地图 (定位导航模式可用) /Switch map (positioning navigation mode is available)
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该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "id": "service_node_manager_control", "values": { "text": "bringup mapping success!", "result": 0 }, "result": true, "service": "/node_manager_control", }</pre>	说明(explain): Op(str): service_response 标示该操作为请求服务的应答 = Op(str): "service_response" indicates that the operation is a reply to the request service. Id(str): 可用于校验对应请求服务的结果 = Id (str): It can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名
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<pre>"op": "service_response" }</pre>	= Service (str): The corresponding service name for the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 Result(bool): Whether the currently requested service is performing correctly. If false.value returns the specific reason for the failure. value: 服务请求对应的返回应答结果,具体内容释义如下 =Value: return answer result corresponding to the service request, defined as follows result(int): 当前指令执行的结果码返回. 错误码释义见下表 /The result code for the current instruction execution is returned. Error code definitions are given in the table below Text(str): 当前指令附加的信息,只是用于 debug 或者 log 输出信息 /The information attached to the current instruction is only for debug or log output information
---------------------------------------	---

错误码对应表如下 /The error codes correspond to the table listed below

错误码 Error code	对应信息 Corresponding information
0	成功操作 /Successful operation
99	未知操作 /Unknown operation
101	程序无法关闭因为根本没启动 /The program cannot be closed because it has not been started at all
102	已经是当前模式 /Already in the current mode
107	输入缺少地图信息 /Enter the missing map information
106	系统致命错误(可能需要重启) /System fatal error (may require restart)
108	地图损坏 /Map damage
109	当前操作不支持(建图状态下不支持切换地图) /Current operation is not supported (switching switching is not supported in mapping state)
110	保存地图失败 /Failed to save map

列出系统楼层地图及概要信息/List the system floor

maps and summary information

该指令的主要功能是用于管理当前系统中保存的地图及相关虚拟墙,标记点信息。

= The main function of this instruction is to manage the maps and related virtual walls saved in the current system, and to mark the point information.

发布该服务对应的请求 (发布)

=The request corresponding to publishing the service (publishing)

<pre>{ "args": { "op": 0 }, "id": "service_layered_map_cmd", "service": "/layered_map_cmd", "op": "call_service" }</pre>	说明(explain): Op(str): call_service 标示该操作为请求服务 /call_service indicates that the operation is a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 /Input parameters corresponding to the service request Json 解析后的数据如下 /The parsed data are presented as follows Building_name(str): 当前地图的建筑名 =Building_name (str): the building name of the current map Floor_name(str): 当前地图的楼层名 =Floor_name (str): The floor name of the current map op(int): 当前对应的操作 =Op (int): the current corresponding action 0: 列出当前系统中所有地图信息 =0: Lists all the map information in the current system
--	--

该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "id": "service_layered_map_cmd", "values": { "info": "list map info success", "list_info": [{ "floor_info": [{ "map": true, "floor_name": "floor_1", "markers": false, "virtual_walls": false }], "building_name": "origin" }, { "floor_info": [{ "map": true, "floor_name": "floor_1", </pre>	说明(explain): Op(str): service_response 标示该操作为请求服务的应答 =Op(str): service_response indicates that the operation is a reply to the request service Id(str): 可用于校验对应请求服务的结果 =Id (str): It can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 =Service (str): The corresponding service name for the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 =Result(bool): Whether the currently requested service is performing correctly. If false.value returns the specific reason for the failure value: 服务请求对应的返回应答结果,具体内容释义如下 =Value: return answer result corresponding to the service
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<pre> "markers": false, "virtual_walls": false }, { "building_name": "\u6211\u7684\u7956\u56fd" }, { "floor_info": [{ "map": true, "floor_name": "floor_4", "markers": true, "virtual_walls": true }, { "map": true, "floor_name": "floor_1", "markers": false, "virtual_walls": false }], "building_name": "cs" }], "result": 0 }, "result": true, "service": "/layered_map_cmd", "op": "service_response" }</pre>	request, defined as follows info(str): 当前指令附加的信息,只是用于 debug 或者 log 输出信息 /The information attached to the current instruction is only for debug or log output information Result(int): 当前返回的结果码,对应的错误编号见下表 /The currently returned result code, the corresponding error number is shown in the table below List_info(str): 当前系统中所有地图及其相关信息,只有当请求对应的 op 设置为 0 时才会返回,具体内容如下 /All maps and their associated information in the current system are returned only when requesting the corresponding op is set to 0. The details are as follows Building_name(str): 建筑名 /Building name Floor_info(list): 当前建筑名下面的所有楼层信息集合 /Collection of all floor information below the current building name Floor_name(str): 楼层名 /Floor name Map(bool): 标识当前是否有地图 /Identifies whether there is a map Virtual_walls(bool): 标识当前是否有虚拟墙信息 /Identifies whether there is currently a virtual wall information Markers(bool): 标识当前是否有标记点位信息 /Identifies whether there is currently a mark point information 当采用 list 的时候 Makers 跟 virtual wall 为空集合,下面会给出另一种列出 markers 跟 virtual wall 形式的返回。 =When list is adopted, Makers and virtual wall are empty sets, and another return of markers and virtual wall is given below.
--	---

删除地图信息 /Delete map information

该指令的主要功能是用于删除当前指定楼层的地图或者整个建筑下所有的地图。

=The main function of this instruction is to delete a map of the currently specified floor or all maps under the entire building.

发布该服务对应的请求 (发布)

=The request corresponding to publishing the service (publishing)

<pre>{ "args": { "floor_name": "123", "op": 0, "building_name": "\u53bb\u53bb\u53bb" }, "id": "service_building_operation/delete", }</pre>	说明(explain): Op(str): call_service 标示该操作为请求服务 /call_service indicates that the operation is a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service
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API development documentation protocol

<pre> "service": "/building_operation/delete", "op": "call_service" } </pre>	name of the request Args(json): 服务请求对应的输入参数 /Input parameters corresponding to the service request Json 解析后的数据如下 /The data resolved by Json are as follows Building_name(str): 当前地图的建筑名 /The building name of the current map Floor_name(str): 当前地图的楼层名 /Floor name of the current map op(int): 当前对应的操作 /The current corresponding action 0: 删除指定的 Building_name 中 Floor_name 的地图 =0: Delete the map for Flower _ name in the specified Building_name 如果 floor_name 为空字符串, 则为删除当前建筑下所有地图 =If the flood _ name is an empty string, delete all maps under the current building
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该服务返回的应答 (接收) /Response returned by this service (receive)

<pre> { "id": "service_building_operation/delete", "values": { "info": "this message success", "list_info": [{ "floor_info": [{ "map": false, "floor_name": "", "markers": false, "virtual_walls": false }], "building_name": "" }], "result": 0, "markers": { "waypoints": [{ "behavior_code": 0, "time_out": 0.0, "pose": { "position": { "y": 0.0, "x": 0.0, "z": 0.0 }, "orientation": { "y": 0.0, </pre>	说明(explain): Op(str): service_response 标示该操作为请求服务的应答 =Op(str): service_response indicates that the operation is a reply to the request service Id(str): 可用于校验对应请求服务的结果 =Id (str): It can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 =Service (str): The corresponding service name for the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 =Result (pool): Whether the currently requested service is performed correctly. For the false.value value: 服务请求对应的返回应答结果,具体内容释义如下 =Value: return answer result corresponding to the service request, defined as follows info(str): 当前指令附加的信息,只是用于 debug 或者 log 输出信息 /The information attached to the current instruction is only for debug or log output information Result(int): 当前返回的结果码,对应的错误编号见下表 /The currently returned result code, the corresponding error number is shown in the table below.
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<pre> "x": 0.0, "z": 0.0, "w": 0.0 } }, "name": "", "rest_time": 0.0 }] }, "vwalls": { "map_name": "", "header": { "stamp": { "secs": 361, "nsecs": 312000000 }, "frame_id": "map", "seq": 0 }, "walls": [{ "end_x": 0.0, "wall_id": "", "end_y": 0.0, "start_x": 0.0, "start_y": 0.0 }] } }, "result": true, "service": "/building_operation/delete", "op": "service_response" } </pre>	
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错误码 Error code	具体意义 Specific significance
0	成功操作 /Successful operation
201	路径无效(参数 map_folder_base 对应, 默认 home/maps) /Invalid path (parameter map_folder_base correspondence, default home/maps)
202	没有对应文件/路径下没有有效地图 No corresponding file / path under no valid map
203	文件内没有数据/文件读取错误 No data / file read errors are present in the file
204	无权限; 删除文件不在 home 目录下 No permission; delete the file is not in the home directory

205	文件损坏; 删除路径失败 File is corrupted; delete path failed
206	获取地图数据超时 /Get map data timeout
207	不知道读取地图(可能在建图状态下) Not known to read map (possibly under state)
208	保存地图失败 /Failed to save map
99	未知指令 /Unknown instructions

获取当前标记点位列表 /Gets the list of current marker points

该指令的主要功能是用于加载当前地图对应的列表记点信息。

=The main function of this instruction is to load the list dot information corresponding to the current map.

发布该服务对应的请求 (发布)

=The request corresponding to publishing the service (publishing)

<pre>{ "args": { "op": 0 }, "id": "service_get_markers", "service": "/marker_operation/get_markers", "op": "call_service" }</pre>	说明(explain): Op(str): call_service 标示该操作为请求服务 =Op (str): call_service indicates that the operation is a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 =Id (str): To set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 =Service (str): The corresponding service name for the request Args(json): 服务请求对应的输入参数 =Args (json):, the input parameters corresponding to the service request Json 解析后的数据如下 =The data resolved by Json are as follows Building_name(str): 当前地图的建筑名 /The building name of the current map Floor_name(str): 当前地图的楼层名 /Floor name of the current map op(int): 当前对应的操作 /The current corresponding action 0: 加载当前地图对应的标记点位信息 =0: Load the marker information corresponding to the current map
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该服务返回的应答 (接收)

=Response returned by this service (receive)

<pre> { "id": "service_marker_operation/get_markers", "values": { "info": "", "list_info": [{ "floor_info": [{ "map": false, "floor_name": "", "markers": false, "virtual_walls": false }], "building_name": "" }], "result": 0, "markers": { "waypoints": [{ "behavior_code": 0, "time_out": 0.0, "pose": { "position": { "y": 0.0, "x": 0.0, "z": 0.0 }, "orientation": { "y": 0.0, "x": 0.0, "z": 0.0, "w": 1.0 } }, "name": "a1", "rest_time": 0.0 }], { "behavior_code": 0, "time_out": 0.0, "pose": { "position": { "y": 3.0, "x": 1.0, "z": 0.0 }, </pre>	说明(explain): Op(str): service_response 标示该操作为请求服务的应答 =Op(str): service_response indicates that the operation is a reply to the request service Id(str): 可用于校验对应请求服务的结果 =Id (str): It can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 =Service (str): The corresponding service name for the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 =Result (pool): Whether the currently requested service is performed correctly. For the false.value value: 服务请求对应的返回应答结果,具体内容释义如下 =Value: return answer result corresponding to the service request, defined as follows info(str): 当前指令附加的信息,只是用于 debug 或者 log 输出信息 /The information attached to the current instruction is only for debug or log output information Result(int): 当前返回的结果码,对应的错误编号见下表 /The currently returned result code, the corresponding error number is shown in the table below 兼容模式消息结构,只需要解析下面部分就好 =Compatible with the mode message structure, just need to parse the following parts Markers.waypoints (list) 为标记点位的列表集合.具体内容释义如下. 接口跟 标记点位信息 中的点位信息格式一致 =Markers.waypoints (list) is a collection of lists of marked points. The specific content is explained as follows. The interface is formatted with the dot information in Marker Point Information Behavior_code(int): 当前标记点的属性,目前 11 为回充点 =Behavior_code (int): the property of the current marker point, 11 is the recharge point Time_out (float): 当前点对应的执行超时时间.单位秒.在规定的时间内如果该点还没有到达则取消该点,执行下一个点位.设置为-1 代表不设置超时时间.一直等待直到该点完成 =Time_out (float): The execution timeout time corresponding to the current point. Unit of seconds. If the point has not arrived within the specified time, cancel the point and execute the next point. Setting to-1 represents a timeout time that is not set. Wait until the point is complete Pose: 当前点位的位置跟角度信息.
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API development documentation protocol

<pre> "orientation": { "y": 0.0, "x": 0.0, "z": 0.0, "w": 1.0 }, "name": "a3", "rest_time": 0.0 } }, "vwalls": { "map_name": "", "header": { "stamp": { "secs": 5543, "nsecs": 187000000 }, "frame_id": "map", "seq": 0 }, "walls": [{ "end_x": 0.0, "wall_id": "", "end_y": 0.0, "start_x": 0.0, "start_y": 0.0 }] } }, "result": true, "service": "/marker_operation/get_markers", "op": "service_response" } </pre>	=Pose: Position and angle information of the current point point. Position.x 地图中对应的 x 坐标轴值 =Position.x The corresponding x-coordinate axis value in the map Position.y 地图中对应的 y 坐标轴值 =Position.y The corresponding y-coordinate axis values in the map Orientation 地图中点位角度对应的四元数 =Orientation The quaternion number corresponding to the point point angle in the map Name (str): 当前点位的对应名称,该名称需唯一.如果不唯一的话其中一个会被覆盖 =Name (str): the corresponding name of the current point point, which should be unique. One of them will be overwritten if it is not unique Rest_time (float) 当前点位的休息时间.到达该点位后等待多少秒后再继续进行去下一个点位 =Rest_time (float) break at the current point. How many seconds do you wait to reach the point before proceeding to the next point
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错误码 Error code	具体意义 Specific significance
0	成功操作 /Successful operation
201	路径无效(参数 map_folder_base 对应, 默认 home/maps) Invalid path (parameter map_folder_base correspondence, default home/maps)
202	没有对应文件/路径下没有有效地图 No corresponding file / path under no valid map

203	文件内没有数据/文件读取错误 No data / file read errors are present in the file
204	无权限；删除文件不在 home 目录下 No permission; delete the file is not in the home directory
205	文件损坏；删除路径失败 File is corrupted; delete path failed
206	获取地图数据超时 /Get map data timeout
207	不知道读取地图(可能在建图状态下) Not known to read map (possibly under state)
208	保存地图失败 /Failed to save map
99	未知指令 /Unknown instructions

删除单个标记点(poi 名称模式) /Remove a single marker point (poi name pattern)

该指令的主要功能是通过设置标记点对应的目标点名称来删除该标记点

=The main function of this instruction is to remove the mark point by setting the corresponding target point name in the mark point

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

<pre>{ "args": { "poi": "p2" }, "id": "service_marker_manager/delete_poi", "service": "/marker_manager/delete_poi", "op": "call_service" }</pre>	说明: explain: Op(str): call_service 标示该操作为请求服务 /Mark the operation as a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 /Input parameters corresponding to the service request Json 解析后的数据如下 /The data resolved by Json are as follows Poi(str): 删除目标点对应的名称 /Delete the name corresponding to the target point
--	---

该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "id": "service_marker_manager/delete_poi", "values": { "available_list": ["a1", "a3", "a2"], "success": true }, "result": true, "service": "/marker_manager/delete_poi", "op": "service_response" }</pre>	说明: explain: Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果 /Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 /Is the currently requested service being performed correctly. For the false.value 如 { "id": "service_XXX", "values": "Service /poi does not exist", "result": false, "service": "/poi", "op": "service_response" } 类似这样表达该服务是因为算法程序端对应的服务没有启动 /The service is expressed like this because the corresponding service on the algorithm program side is not started value: 服务请求对应的返回应答结果,具体内容释义如下 /The service request corresponds to the return of the answer result, the specific content is defined as follows Success(bool): 当前目标点是否成功发布.如果为 false 则代表目前对应的点位列表中没有该名称的对应点.可参照 available_list 检验当前可用点 /Whether the current target point was successfully published. If false, it represents the corresponding point without that name in the current corresponding point list. Refer to the currently available points in the available_list test Available_list(list), 字符串格式的列表数组,包含了当前对应地图中设置的标记点位 /A list array in string format that contains the marker points set in the current corresponding map
--	---

发布导航点信息(poi 名称模式)/Publish navigation point information (poi name mode)

该指令的主要功能是通过设置标记点对应的目标点名称来进行单点导航

=The main function of this instruction is to perform a single point navigation by setting the corresponding target point name in the tag point

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

{	说明: explain:
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API development documentation protocol

<pre> "args": { "poi": "P1" }, "id": "service_poi", "service": "/poi", "op": "call_service" } </pre>	Op(str): call_service 标示该操作为请求服务 /Mark the operation as a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 /Input parameters corresponding to the service request Json 解析后的数据如下 /The parsed data are presented as follows Poi(str): 发布目标点对应的名称 /The name corresponding to the publication target point
--	--

该服务返回的应答（接收） /Response returned by this service (receive)

<pre> { "id": "service_poi", "values": { "available_list": ["P2", "P3", "P0", "P1"], "success": true }, "result": true, "service": "/poi", "op": "service_response" } </pre>	说明: explain: Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果 /Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 /Is the currently requested service being performed correctly. Return the specific reason for the failure for false.value 如 {"id": "service_XXX", "values": "Service /poi does not exist", "result": false, "service": "/poi", "op": "service_response"} 类似这样表达该服务是因为算法程序端对应的服务没有启动 /The service is expressed like this because the corresponding service on the algorithm program side is not started value: 服务请求对应的返回应答结果,具体内容释义如下 /The service request corresponds to the return of the answer result, the specific content is defined as follows Success(bool): 当前目标点是否成功发布.如果为 false 则代表目前对应的点位列表中没有该名称的对应点.可参照 available_list 检验当前可用点 /Whether the current target point was successfully published. If false, it represents the corresponding point without that name in the current corresponding point list. Refer to the currently available points in the available_list test Available_list(list), 字符串格式的列表数组,包含了当前对应地图中设置的标记点位 /A list array in string format that contains
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	the marker points set in the current corresponding map
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获取当前运行中地图信息接口

A list array in string format that contains the marker points set in the current corresponding map

该指令的主要功能是获取当前地图的信息

=The main function of this instruction is to obtain information about the current map.

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

<pre>{ "args": { "cmd": 0 }, "id": "service_get_map_info", "service": "/get_map_info", "op": "call_service" }</pre>	说明: explain: Op(str): call_service 标示该操作为请求服务 /Mark the operation as a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 /Input parameters corresponding to the service request Json 解析后的数据如下 /The parsed data are presented as follows Cmd(int): 0: 获取数据 /getting data
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该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "id": "service_get_map_info", "values": { "height": 256, "width": 352, "origin_y": -9.6000003815, "origin_x": -9.6000003815, "free_space": 200.0, "occ_space": 15.5, } }</pre>	说明: explain: Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果 /Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 /Is the currently requested service being
---	---

<pre>"unknown_space": 10.28, "floor_name": "1", "resolution": 0.0500000007, "building_name": "tefa" }, "result": true, "service": "/get_map_info", "op": "service_response" }</pre>	performed correctly. Return the specific reason for the failure for false.value value: 服务请求对应的返回应答结果,具体内容释义如下 /The service request corresponds to the return of the answer result, the specific content is defined as follows Origin_x(double) 第一个像素值所代表的全局坐标值 x 轴坐标 /The service request corresponding to the response result, the specific content is as follows Origin_x (double) The global coordinate value x-axis coordinate represented by the first pixel value Origin_y(double) 第一个像素值所代表的全局坐标值 y 轴坐标 /The global coordinate value y-axis coordinate represented by the first pixel value Width (int): 地图的宽 /The width of the map Height(int): 地图的高 /The height of the map Resolution (double) 地图的分辨率, 一个像素代表实际大小, 示例 0.05m 一个像素格 /The resolution of the map, one pixel represents the actual size, example 0.05m a pixel lattice building_name(str) 当前地图对应的建筑名 /The building name corresponding to the current map floor_name(str) 当前地图对应的楼层名 /The floor name corresponding to the current map free_space(double): 当前地图空白区域面积 (m²) /Current map blank area area (m2) occ_space(double): 当前地图占用区域面积 (m²) /Current map occupied area (m2) unknown_space(double): 当前地图未知区域面积 (m²) /Current map unknown area area (m2) 探索区域面积 = 空白区域面积 + 占用区域面积 =Exploration area = blank area + occupied area
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导航速度修改/Navigation speed modification

该指令的主要功能是用于设置导航时所需要执行的速度模式

=The main function of this instruction is to set the speed mode that needs to execute during navigation

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

{	说明: explain:
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API development documentation protocol

<pre> "args": { "cmd": 3, "str": "" }, "id": "service_velocity_control", "service": "/velocity_control", "op": "call_service" } </pre>	Op(str): call_service 标示该操作为请求服务 /Mark the operation as a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 Json 解析后的数据如下 Cmd(int): 速度模式, 分为高中低三种速度档位。每个档位配有 安全、平衡、效率三种模式。总共组合有九种可能性 /The resolved data of the input parameter Json corresponding to the service request is as follows Cmd (int): speed mode, divided into three high, high and low speed gears. Each gear has three modes of safety, balance and efficiency. There are nine possibilities for the total combinations 0: 安全低速 /Safety and low speed 1: 安全中速 /Safety and low speed 2: 安全高速 /Safety and high speed 3: 平衡低速 /Balance low speed 4: 平衡中速 /Balance medium speed 5: 平衡高速 /Balance high speed 6: 效率低速 /Low efficiency 7: 效率中速 /Efficiency medium speed 8: 效率高速 /High efficiency -1: 默认模式 (出厂默认参数), 此参数只为读取过渡。设置后下次开机后生效。也可以认为该模式为平衡中速 /Default mode (factory default parameter), which is only for read transition. It will take effect after the next startup after setting. This mode can also be considered as an equilibrium medium velocity 60: 平稳模式 (送餐模式), 开启平稳模式 /Smooth mode (food delivery mode), open the smooth mode 61: 关闭平稳模式 (送餐模式), 关闭平稳模式速度会将速度设置设置为默认模式 -1 /Turn off smooth mode (food delivery mode) and turning off smooth mode speed sets the speed setting to default mode-1 99: 获取当前速度模式 /Get the current speed mode Str(str):
--	--

该服务返回的应答 (接收) /Response returned by this service (receive)

{	说明: explain:
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<pre> "info": "this message success", "service": "/velocity_control", "values": { "info": "balance mode with low speed", "result": 0 }, "result": true, "id": "service_velocity_control", "op": "service_response" } </pre>	Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果 /Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 /Whether the currently requested service is performing correctly. If false.value returns the specific reason for the failure value: 服务请求对应的返回应答结果,具体内容释义如下 /The service request corresponds to the return of the answer result, the specific content is defined as follows info(str): 当前指令附加的信息,只是用于 debug 或者 log 输出信息。在输入为获取当前速度模式 (cmd=99) 情况下, 返回字符串为数字型字符串内容是当前系统对应的速度模式的值 /The information attached to the current instruction is only for debug or log output information. When the input is to obtain the current speed mode (cmd = 99), the return string is digital and the string content is the value of the speed mode corresponding to the current system Result(int): 返回结果码, 0 为成功执行, 99 为未知指令 /Return result code, 0 for successful execution and 99 for unknown instruction 620: 对应速度配置文件未找到 /The corresponding speed profile is not found 621: 动态修改速度失败, 具体失败原因见 info 字段 /Dynamic modification speed failed, see the info field
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硬件检测 /hardware test

该指令的主要功能是用于底盘传感器的自检，用于检测配置传感器是否正常工作。

=The main function of this instruction is for self-inspection of the chassis sensor to detect the configuration sensor.

服务调用后需要等待一段时间程序调用各个传感器数据来检测是否正常工作

=After the service call, you need to wait a period of time for the program to call each sensor data to detect whether it works properly.

目前检测传感器有：编码器左/右，电池电压，充电器连接，超声波左/中/右，激光雷达，深度相，网络连接，陀螺仪，红外回充信号

=At present, the detected sensors are: encoder left / right, battery voltage, charger connection,

ultrasonic left / middle / right, lidar, deep phase, network connection, gyroscope, infrared recharge signal.

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

<pre>{ "args": {}, "id": "service_self_diagnosis", "service": "/self_diagnosis", "op": "call_service" }</pre>	说明: explain: Op(str): call_service 标示该操作为请求服务 /Mark the operation as a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 无附加输入参数 /No additional input parameters are available
---	--

该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "info": "this message success", "service": "\self_diagnosis", "values": { "status": [{ "message": "激光雷达正常工作", "hardware_id": "激光雷达", "values": [], "name": "", "level": 0 }, { "message": "网络连接正常", "hardware_id": "网络连接", "values": [], "name": "", "level": 0 }, { "message": "编码器左 传感器数据未收到", "hardware_id": "编码器左", "values": [], "name": "", "level": 2 }, { "message": "编码器右 传感器数据未收到", "hardware_id": "编码器右", "values": [], "name": "", "level": 2 }] }</pre>	说明: explain: Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果 /Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 /Whether the currently requested service is performing correctly. If false.value returns the specific reason for the failure value: 服务请求对应的返回应答结果,具体内容释义如下 /The service request corresponds to the return of the answer result, the specific content is defined as follows Passed(byte): 0 无需理会 /No need to pay attention Id(string): " 无需理会 / No need to pay attention Status (list) 列表型数组数据, 多个传感器状态的集合。数组元素结构如下 /List of array data, a collection of multiple sensor states. The array elements are structured as follows Hardware_id(str): 传感器设备的名称 /The name of the sensor device Level (int): 当前设备的状态等级 /The status level of the current device 0: 正常工作 /regular work 1: 警告/warn 2: 错误/wrong
--	---

<pre>数据未收到", "hardware_id": "编码器右", "values": [], "name": "", "level": 2 }, { "message": "电池电压 传感器 数据未收到", "hardware_id": "电池电压", "values": [], "name": "", "level": 2 }, { "message": "充电器连接 传感 器数据未收到", "hardware_id": "充电器连接", "values": [], "name": "", "level": 2 }, { "message": "红外信号数据未收 到", "hardware_id": "红外回充信号", "values": [], "name": "", "level": 2 }, { "message": "超声波左 传感器 数据未收到", "hardware_id": "超声波左", "values": [], "name": "", "level": 2 }, { "message": "超声波中 传感器 数据未收到", "hardware_id": "超声波中", "values": [], "name": "", "level": 2 }, { "message": "超声波右 传感器 数据未收到", "hardware_id": "超声波右", "values": [],</pre>	3: 数据超时/Data timeout Message (str) : 附加输出信息/Additional output information Name (str): 输出具体错误码/Output the specific error code
---	--

```
        "name": "",
        "level": 2
    }, {
        "message": "陀螺仪数据正常",
        "hardware_id": "陀螺仪",
        "values": [],
        "name": "",
        "level": 0
    }, {
        "message": "深度相机正常",
        "hardware_id": "深度相机",
        "values": [],
        "name": "",
        "level": 0
    }
},
"id": "",
"passed": 0
},
"result": true,
"id": "service_self_diagnosis",
"op": "service_response"
}
```

机器人版本信息/Robot version information

该指令的主要功能是用于获取机器人软硬件版本信息

=The main function of this instruction is to obtain information about the hardware version of the robot software

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

```
{
  "args": {
    "cmd": 0
  },
  "id": "service_robot_info",
  "service": "/robot_info",
  "op": "call_service"
}
```

说明: explain:

Op(str): call_service 标示该操作为请求服务 **/Mark the operation as a requested service**

Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 **/Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal**

Service(str): 请求对应的服务名 **/The corresponding service name of the request**

Args(json): 服务请求对应的输入参数 Json 解析后的数据如下 **/The resolved data of the input parameter Json corresponding to the service request is as follows**

	Cmd(int): 版本相关指令, 释义如下 /Version related instructions, the definition is as follows 0: 获取当前机器人底盘的相关信息 /Obtain relevant information about the current robot chassis
--	---

该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "info": "this message success", "service": "/robot_info", "values": { "robot_id": "TY1251C003_0036", "firmware_version": "1.0.2", "robot_type": "TONG", "software_version": "4.1.0-alpha", "hardware_version": "2.0.0" }, "result": true, "id": "service_robot_info", "op": "service_response" }</pre>	说明: explain: Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果 /Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行. 如果为 false, value 会返回具体失败原因 /Is the currently requested service being performed correctly. Return the specific reason for the failure for false.value value: 服务请求对应的返回应答结果, 具体内容释义如下 /The service request corresponds to the return of the answer result, the specific content is defined as follows Robot_id(str): 机器人编号 /Robot number Robot_type(str): 机器人类型 /Robot type Software_version(str): 算法版本 /Algorithm version firmware_version(str): 固件版本 /Firmware version hardware_version(str): 硬件版本 /Hardware version
---	---

获取当前地图匹配阈值 /Take the current map matching threshold

该指令的主要功能是用于获取当前系统中设置的地图匹配的阈值.

=The main function of this instruction is to obtain the threshold for map matching set in the current system.

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

<pre>{ "args": { "cmd": 0, "str": "" }, "id": "service_get_match_threshold", "service": "/get_match_threshold", "op": "call_service" }</pre>	说明: explain: Op(str): call_service 标示该操作为请求服务 /Mark the operation as a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 Json 解析后的数据如下 /The resolved data of the input parameter Json corresponding to the service request is as follows Cmd(int): 版本相关指令, 释义如下 /Version related instructions, the definition is as follows 0: 获取当前系统设置的地图匹配阈值 /Gets the map matching threshold set by the current system
--	---

该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "info": "this message success", "service": "/get_match_threshold", "values": { "info": "setting match threshold to 80 %", "result": 80 }, "result": true, "id": "service_get_match_threshold", "op": "service_response" }</pre>	说明: explain: Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果 /Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value 会返回具体失败原因 /Is the currently requested service being performed correctly. Return the specific reason for the failure for false.value value: 服务请求对应的返回应答结果,具体内容释义如下 /The service request corresponds to the return of the answer result, the specific content is defined as follows Result(int): 返回阈值.直接为百分比数值 /Returns the threshold value. Directly to the percentage value
--	---

计算点位之间距离/Calculate the distance between the point points

该指令的主要功能是用于计算当前地图中两个点位之间的距离.该距离是根据地图信息计算出来的路径规划出来的导航距离

=The main function of this instruction is to calculate the distance between the two points in the current map. This distance is the navigation distance planned by the path calculated from the map information.

发布该服务对应的请求 (发布)/The request corresponding to publishing the service (publishing)

<pre>{ "args": { "goal_x": 1.1, "goal_y": 1.2, "start_x": 0.1, "start_y": 0.2, "start_floor": "1", "goal_floor": "1" }, "id": "service_calculate_distance", "service": "/calculate_distance", "op": "call_service" }</pre>	说明: explain: Op(str): call_service 标示该操作为请求服务 /Mark the operation as a requested service Id(str): 用于设置该服务对应的 id 号.可用于判断应答信号对应的返回值 /Use to set the corresponding id number for the service. Can be used to determine the return value corresponding to the response signal Service(str): 请求对应的服务名 /The corresponding service name of the request Args(json): 服务请求对应的输入参数 Json 解析后的数据如下 /The resolved data of the input parameter Json corresponding to the service request is as follows start_x (double): 起点 x 轴坐标 /Starting point x axis coordinates start_y (double): 起点 y 轴坐标/Start y-axis coordinates start_floor (str): 起点所在楼层/Floor where the starting point is goal_x (double): 终点 x 轴坐标/End point x axis coordinates goal_y (double): 终点 y 轴坐标/End point y-axis coordinates goal_floor (str): 终点所在楼层/Floor where the end point is
--	--

该服务返回的应答 (接收) /Response returned by this service (receive)

<pre>{ "info": "this message success", "service": "/calculate_distance", "values": { "info": "", "distance": 1.4309386487, "result": 0 }, }</pre>	说明: explain: Op(str): service_response 标示该操作为请求服务的应答 /A response marking that the operation is a requested service Id(str): 可用于校验对应请求服务的结果/Can be used to verify the results of the corresponding requested service Service(str): 请求对应的服务名 /The corresponding service name of the request Result(bool): 当前请求的服务是否正确执行.如果为 false.value
---	---

<pre>"result": true, "id": "service_calculate_distance", "op": "service_response" }</pre>	会返回具体失败原因/Is the currently requested service being performed correctly. Return the specific reason for the failure for false.value value: 服务请求对应的返回应答结果,具体内容释义如下/The service request corresponds to the return of the answer result, the specific content is defined as follows info(str): 当前指令附加的信息,只是用于 debug 或者 log 输出信息/The information attached to the current instruction is only for debug or log output information Result(int): 返回结果码, 0 为成功执行, 10 为上次指令在运行中.151 为路径计算服务器出错. 150 为路径规划失败,地点不可到达. 152 为不支持跨楼层计算./Return the result code, 0 is the successful execution, 10 is the last instruction in operation. 151 Error occurred for the path calculation server. 150 For path planning failure, the location is not reach. 152 Cross floor calculations is not supported. Distance(double): 返回路径距离值.单位米./Return the path distance value. Unit of meter. Info(str): 附加信息, 在返回结果码不成功的时候包含附加信息用于调试 /Additional information, including additional information for debugging when the return result code is unsuccessful
---	---

备注/remark

状态码表/Status code table

通用标准输出状态/# General Standard output status

0: 指令成功执行/The instructions were executed successfully

10: 指令正在处理中/The instruction is being processed

20: 正在建图状态/Building diagram state

30: 正在定位导航状态/Locating the navigation status

99: 未知的输入指令/Locating the navigation status

程序控制框架组# Program control frame group

101: 关闭程序出错,因为根本没启动/Error in closing the program because it was not started at all

102: 启动程序无效,因为程序已经启动了/The startup program is invalid because the program is already started

104: 没有对应地图/There is no corresponding map

105: 没有权限控制程序/No permission control program

106: 系统致命错误,请重启/System fatal error, please restart

107: 缺失建筑或楼层信息/Missing building or floor information

108: 地图文件损坏/The map file is damaged

109: 当前模式不支持该指令/The directive is not supported by the current mode

110: 保存地图错误/Save map error

楼层管理组 2XX, 地图管理组 3XX, 虚拟墙组 4XX, 标记点位组 50X. 区域管理 55X

地图管理,虚拟墙,标记点位是在楼层管理组状态码基础上百位的值不一样

Floor management group 2XX, map management group 3XX, virtual wall group 4XX, marked point point group 50X., area management 55X

Map management, virtual wall, marking points are in the floor management group status code based hundreds of different values

201: 路径无效/The path is invalid

202: 没有对应文件/There is no corresponding file

203: 没有数据内容/There is no data content

204: 没有权限访问/There is no permission to access

205: 文件损坏/File damage

206: 地图数据获取超时/Map data acquisition timeout

207: 建图模式不支持加载地图/Loading map is not supported for mapping mode

208: 保存数据文件失败/Failed to save the data file

导航相关/# Navigation related

600: 初始化等待/Initialize waiting

601: 运行中/in service

602: 导航取消/Navigation cancelled

603: 导航成功/Navigate successfully

604: 导航失败/Navigation failed

605: 导航拒绝/Navigation refused

606: 导航取消中 In navigation cancellation

607: 导航重置中/Navigation reset

608: 导航重置完成/Navigation reset completed

609: 导航状态丢失/The navigation status is lost

620: 配置失败/Configuration failed

621: 急停按钮触发/Urgent stop button trigger

跨楼层导航相关,需梯控配合/# Cross-floor navigation is related, need ladder control cooperation

630: 跨楼层导航输入错误/Wrong navigation input across floors

631: 跨楼层导航目标楼层无效/Navigating the target floor across floors is invalid

632: 跨楼层导航目标标记点无效/Cross-floor navigation target markers are invalid

630: 跨楼层导航输入错误/Wrong navigation input across floors

631: 跨楼层导航目标楼层无效/Navigating the target floor across floors is invalid

632: 跨楼层导航目标标记点无效/Cross-floor navigation target markers are invalid

633: 跨楼层导航获取标记点位错误/Cross-floor navigation gets the mark point error

634: 跨楼层导航获取电梯相关点位无效/Cross-floor navigation to obtain the elevator related points is invalid

635: 跨楼层导航失败/Navigation across floors failed

636: 跨楼层导航取消/Cross-floor navigation is cancelled

637: 跨楼层导航预约电梯失败/Cross-floor navigation and elevator reservation appointments have failed

639: 跨楼层导航等待中/Cross-floor navigation is waiting for

640: 跨楼层导航检测可行性/Cross-floor navigation detection feasibility

641: 导航到电梯外/Navigate outside the elevator

642: 等待电梯到达指定楼层/Wait for the elevator to reach the designated floor

643: 进入电梯中/Enter the elevator

644: 切换地图中/Switch in the map

645: 离开电梯中/Leave the elevator

646: 导航到目标点/Navigate to target point

647: 跨楼层导航完成/Cross-floor navigation is complete

多点导航/# Multi-point navigation

650: 多点导航完成/Multi-point navigation completed

651: 多点导航运行中/Multi-point navigation is in operation

652: 多点导航暂停/Multi-point navigation is in operation

653: 多点导航继续/Multi-point navigation continues

656: 多点导航可以运行,但是部分点位无效/Multi-point navigation can be run, but some points are not valid

657: 只有单个点有效,不够点位进行多点导航/Only a single point is valid, not enough points for multi-point navigation

658: 多点导航在运行中/Multipoint navigation is in the running

659: 多点导航没有在运行中/Multipoint navigation is not in the running

660: 多点导航没有足够点进行巡航/Multi-point navigation is not enough to cruise

版本升级相关/# Version upgrade is related

701: 没有找到 U 盘/No U disk found

702: U 盘没有找到算法更新包/The U disk did not find the algorithm update package

703: U 盘中非有效更新包/Non-valid update package in the U disk

704: 更新包损坏/Update package is damaged

711: 没有网络/lack of network; no network; without network

712: 下载更新包配置文件失败/Failed to download the update package profile

713: 下载更新包失败/Failed to download the update package

714: 计算更新包校验和失败/Calculate update package checksum failed

715: 更新包校验和不通过/Update package checksum failed

716: 更新服务器不在线/The update server is not online

717: 固件包查找失败/Firmware package search failed

762: 获取网络时间失败/Failed to get the network time

763: 设置系统时间失败/Failed to set the system time

788: 更新包解压失败/Update pack decompression failed

红外回充相关/# IR recharge correlation

901: 回充成功/Recharging success

902: 回充无法检测到红外信号/The infrared signal could not be detected by the charge recharge

903: 无法找到充电桩/Charging pile could not be found

904: 回充超时/Recharging timeout

905: 回充运行中/Recharging operation

906: 回充取消/Recharging cancellation

907: 回充后退中检测到障碍物/An obstacle is detected in the backward recharge process

908: 回充等待/Recharging waiting

910: 回充导航到充电桩前失败/Back to charge the navigation to the charging pile failed

912: 后方有障碍物导致回充电桩失败/An obstacle in the rear leads to the failure of the return charging pile

913: 红外信号没有检测到/The infrared signal was not detected

914: 红外回充无检测信号/No detection signal for infrared recharge

915: 回充电桩上超时/Timeout on the return charging pile

916: 回充电桩前/Back to the charging pile

917: 寻找充电桩信号/Look for the charging pile signal

918: 回充导航到充电桩时取消/Cancel during the recharge and navigation to the charging pile

919: 回充等待中/Recharging waiting

920: 回充再次尝试/Recharging tried again

全局自定位相关/# Global self-localization correlation

1001: 自定位正在运行中/Self-positioning is running

1002: 区域自定位输入太少顶点信息/Area self-localization to input with too little vertex information

1003: 指定位姿搜索输入信息有误,非单个点位表达/The specified position search input information is wrong, not a single point expression

1004: 非定位模式下,自定位不可用/In nonpositioning mode, self-positioning is not available

1005: 定位时雷达数据太少或者无效/Too little or invalid radar data during positioning

1099: 自定位意外失败,需要日志排查/Self-positioning accidentally failed, requiring log screening

主动通知内容以及字段内容/Active notification content and field content

name	function	level	topics	msg	功能备注/Function remarks
system_monitor 系统传感器检测异常上报/System sensor detection abnormal report	激光雷达/ladar	16	无/	错误具体信息/Error specific information	激光雷达异常/Laser radar anomaly
	底盘驱动数据 Chassis drive data	16	错误传感器名称数组/Error-sensor name array	传感器数据对应错误码, ; 号分隔/Sensor data corresponds to error code; separated	底盘传感器异常. 但各个模块采用拼接字符串形式上传. 主要是左电机, 右电机, IMU 以及电池/Chassis sensor is abnormal. However, each module is uploaded in the form of a splicing string. Mainly left motor, right motor, IMU and battery
safety_controller 安全防护相关/Safety protection related	laser_match	4	/	数据内容中包含该当前匹配度值/The current match degree value is included in the data content	当前地图匹配阈值低/Current map matching threshold is low
		1	/	提示功能开关/Tip function switch	地图匹配功能开关提示/Map matching function switch prompt
	laser_data	4	/	提示雷达大部分数据无效/Suggests that most of radar data is invalid	当前大部分激光雷达数据无效/Most of the current lidar data is invalid

		1/	提示功能开关/Tip function switch	激光数据有效性开 关提示/Laser data validity switch prompt
--	--	----	-------------------------------	---

相关算法知识/Related algorithm knowledge

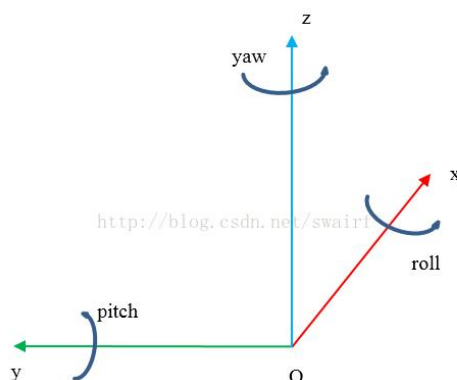
系统坐标系/System coordinate system

3D 笛卡尔坐标系（xyz 坐标系），这里采用右手坐标系。也就是右手的中指为 x 轴，食指为 z 轴，大拇指为 y 轴。并规定了 x 轴为正前方。

=3D Cartesian coordinate system (xyz coordinate system), here using the right-hand coordinate system. That is, the middle finger of the right hand is x axis, the index finger is z axis, and the thumb is y axis. And the x-axis is set right in front.

还有三个欧拉角度来定义绕着这三个轴旋转的角度，绕着 x 轴的为 roll，y 轴为 pitch，z 轴为 yaw。并且逆时针为正方向

=There are also three Euler angles to define the rotation around these three axes, with the roll around the x-axis, the y-axis being the patch, and the z-axis being the yaw. And counterclockwise.



雷达点云转换到全局坐标系下/The radar point cloud is converted to the global coordinate system

雷达点云数据的表达是在机器人坐标系下的表达，并不能直接绘制在地图上显示。所以需要

进行一个转换到全局坐标系下的表达步骤。之所以在这里加这一个步骤而不是直接输出成在全局坐标系下，因为这样点云数据相对与底盘的位置信息。可以方便上层用户根据点云信息做其他处理。如果直接输出在地图下的表达，这样会导致无法直接使用点云数据，转换成原始点云数据反而更麻烦

=The expression of radar point cloud data is expressed in the robot coordinate system and cannot be displayed directly displayed on the map. So you need to do an expression step that transforms to the global coordinate system. The reason why this step is added here rather than directly output into the global coordinate system is, because so that the point cloud data is relative to the position information of the chassis. It is convenient for the upper-level users to do other processing according to the point cloud information. If the expression under the map, it will make it impossible to use the point cloud data directly, making it more troublesome to convert into the original point cloud data.

首先转换到全局坐标系下，首先需要知道当前机器人位置 (xRP, yRP, heading)

=First switch to the global coordinate system, first need to know the current robot position (xRP, yRP, heading)

xRP, yRP 机器人在全局坐标系下的坐标， heading 是机器人当前的方向，单位弧度

=The xRP, the coordinate of the yRP robot in the global coordinate system, heading is the current direction of the robot, the unit radian.

x_laser 是雷达点云列表中对应的 x 轴坐标系

=The x_laser is the corresponding x-axis coordinate system in the radar point cloud list.

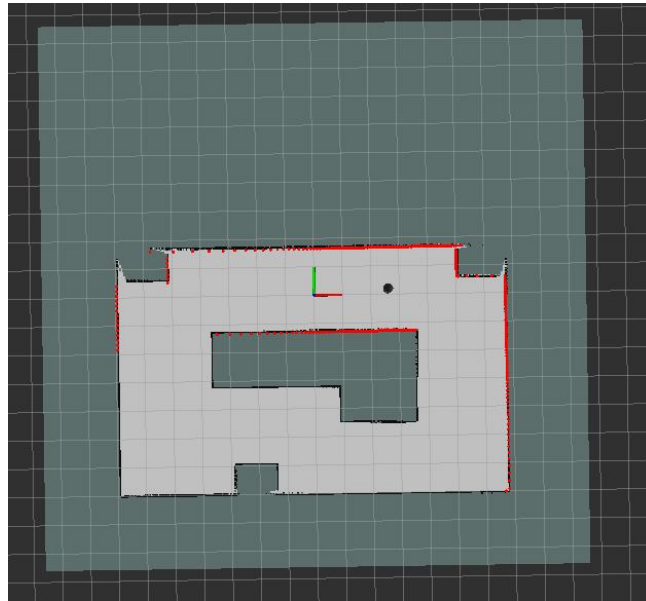
y_laser 是雷达点云列表中对应的 y 轴坐标系

=The y_laser is the corresponding y-axis coordinate system in the radar point cloud list.

xw, yw 是雷达点云转换到全局坐标系下的坐标列表

The xw, yw is the list of coordinates where the radar point cloud transforms to the global coordinate system.

```
for (int i=0; i<x_laser.size(); i++)
{
    xw[i] = xRP + (x_laser[i] * cos(heading)) - (y_laser[i] * sin(heading));
    yw[i] = yRP + (x_laser[i] * sin(heading)) + (y_laser[i] * cos(heading));
}
```



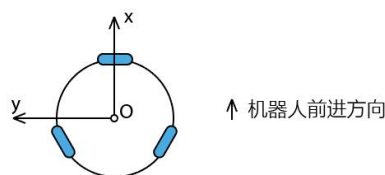
红色点云数据，黑色标记物是机器人位置。理想匹配正确情况下，这两个组合好后应该是能跟实际地图所对应重合

=Red dot cloud data, and black markers are robot locations. If the ideal match is correct, the two combinations should be able to coincide with the actual map.

差分驱动速度控制/Differential drive speed control

针对机器人本体的运动控制也是按照右手坐标系。由于是差分驱动的方式控制机器人，所以只有线速度 (v_x) 跟角速度 (w_z) 的构成。也就是沿着 x 轴直线前进的速度跟绕着 z 轴旋转 (yaw) 的角速度

=The motion control for the robot body is also controlled according to the right-hand coordinate system. Since the robot is controlled by differential drive, only the linear speed (v_x) and the angular speed (w_z) are formed. That is, the speed of moving straight forward along the x -axis follows the angular speed of the rotation (yaw) around the z -axis.



欧拉角与四元数转换/Euler angle with a quaternion transformation

在 3D 图形学中，最常用的旋转表示方法便是四元数和欧拉角，比起矩阵来具有节省存储空间和方便插值的优点。本文主要归纳了两种表达方式的转换，计算公式采用 3D 笛卡尔坐标系：

In 3D graphics, the most commonly used rotation representation methods are the quaternion and Euler angle, which have the advantages of saving storage space and convenient interpolation compared with the matrix. This paper mainly summarizes the transformation of two expressions, and the calculation formula adopts 3D Cartesian coordinate system:

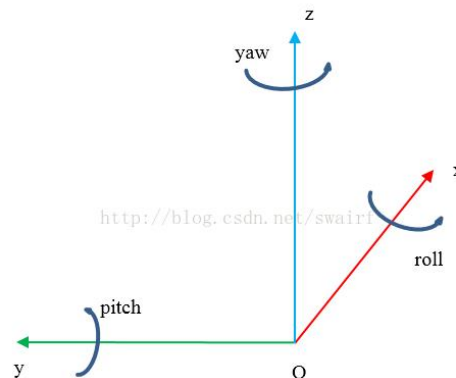


图 1 右手坐标系系统 Figure 1 Right-hand coordinate system system

定义 yaw (ψ), pitch (θ), roll (ϕ) 分别为绕 Z 轴、Y 轴、X 轴的旋转角度

Define yaw (ψ), pitch (θ), and roll (ϕ) as the rotation angles around the Z axis, Y axis, and X axis, respectively

欧拉角到四元数的转换/Euler angle to quaternionic conversion

$$q = \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos(\phi/2)\cos(\theta/2)\cos(\psi/2) + \sin(\phi/2)\sin(\theta/2)\sin(\psi/2) \\ \sin(\phi/2)\cos(\theta/2)\cos(\psi/2) - \cos(\phi/2)\sin(\theta/2)\sin(\psi/2) \\ \cos(\phi/2)\sin(\theta/2)\cos(\psi/2) + \sin(\phi/2)\cos(\theta/2)\sin(\psi/2) \\ \cos(\phi/2)\cos(\theta/2)\sin(\psi/2) - \sin(\phi/2)\sin(\theta/2)\cos(\psi/2) \end{bmatrix}$$

四元数到欧拉角的转换/Conversion of the quaternions to the Euler angle

$$\begin{bmatrix} \varphi \\ \theta \\ \psi \end{bmatrix} = \begin{bmatrix} \arctan \frac{2(wx + yz)}{1 - 2(x^2 + y^2)} \\ \arcsin(2(wy - zx)) \\ \arctan \frac{2(wz + xy)}{1 - 2(y^2 + z^2)} \end{bmatrix}$$

$\arctan$ 和 $\arcsin$ 的结果是 $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ ，这并不能覆盖所有朝向(对于 θ 角 $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ 的取值范围已经满足)，因此需要用 atan2 来代替 $\arctan$ 。

= The result of Arctan and ArcSIN is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, which does not cover all orientations (the range

of values for θ angle $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is satisfied), so atan2 needs to be used instead of $\arctan$.

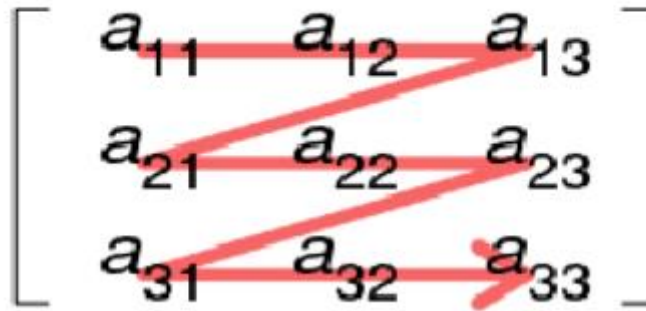
$$\begin{bmatrix} \varphi \\ \theta \\ \psi \end{bmatrix} = \begin{bmatrix} \text{atan2}(2(wx + yz), 1 - 2(x^2 + y^2)) \\ \arcsin(2(wy - zx)) \\ \text{atan2}(2(wz + xy), 1 - 2(y^2 + z^2)) \end{bmatrix}$$

地图的像素坐标与世界坐标转换/The pixel coordinates of the map are transformed to the world coordinates

地图的存储格式为 row-major order 的方式进行存储。如下图所示

=Maps are stored in a row-major order mode for storage format. As shown in the figure below.

Row-major order



Row-major order,
e.g., C (0-indexed)

Address	Access	Value
0	A[0][0]	a_{11}
1	A[0][1]	a_{12}
2	A[0][2]	a_{13}
3	A[1][0]	a_{21}
4	A[1][1]	a_{22}
5	A[1][2]	a_{23}

也就是当解析图片信息数据的时候，地图真实数据的存储是采用列表的形式进行存储。在读取解析的时候采用 row-major order。如第三个地图数据列表中的元素代表着第[0][2] / a_{13} 的数据

=That is, when the image information data, the storage of map real data is stored in the form of list. Use row-major order when the read is parsed. As the elements in the third map data list represent [0][2] / a_{13}

像素坐标的表达就是图像中像素所在的行列值，这个是个整型值

=The expression of the pixel coordinates is the ranks of the pixels in the image, which is an integer value.

世界坐标的表达就是在一个固定坐标系下的表达，一般会给定图片中第[0][0]个像素的真实位置，是个浮点数的坐标值，再根据地图的分辨率大小跟图片的长宽进行运算解析

=The expression of the world coordinate is the expression in a fixed coordinate system. Generally, the real position of the [0][0] th pixel in the picture is given as the coordinate value of a floating point number, and then the calculation is analyzed according to the resolution size of the map and the length and width of the picture.

需要注意的是，根据不同的编程软件绘图，图片的显示可能不太一样。一般而言，地图的第

一个像素是在绘制图片的左上角。而人观测的时候喜欢是在左下角。所以可能需要做一个对称翻转

=It is important to note that the picture may not be displayed differently according to the different programming software drawings. In general, the first pixel of a map is in the upper left corner of the drawing picture. And people like to watch it in the lower left corner. So you might need to do a symmetry flip.

像素坐标转换成世界坐标 /The pixel coordinates are converted into the world coordinates

像素坐标 pixel coordinates (px, py)

世界坐标 world coordinate (wx,wy)

origin_x, origin_y 为地图中的第一个值对应的世界坐标值 /origin_x, origin_y is the world coordinate value for the first value in the map

$$\begin{aligned}wx &= \text{origin_x} + \text{px} * \text{resolution} \\wy &= \text{origin_y} + \text{py} * \text{resolution}\end{aligned}$$

世界坐标转换成像素坐标 /The world coordinates are converted into the pixel coordinates

$$\begin{aligned}\text{px} &= (\text{wx} - \text{origin_x}) / \text{resolution} \\\text{py} &= (\text{wy} - \text{origin_y}) / \text{resolution}\end{aligned}$$